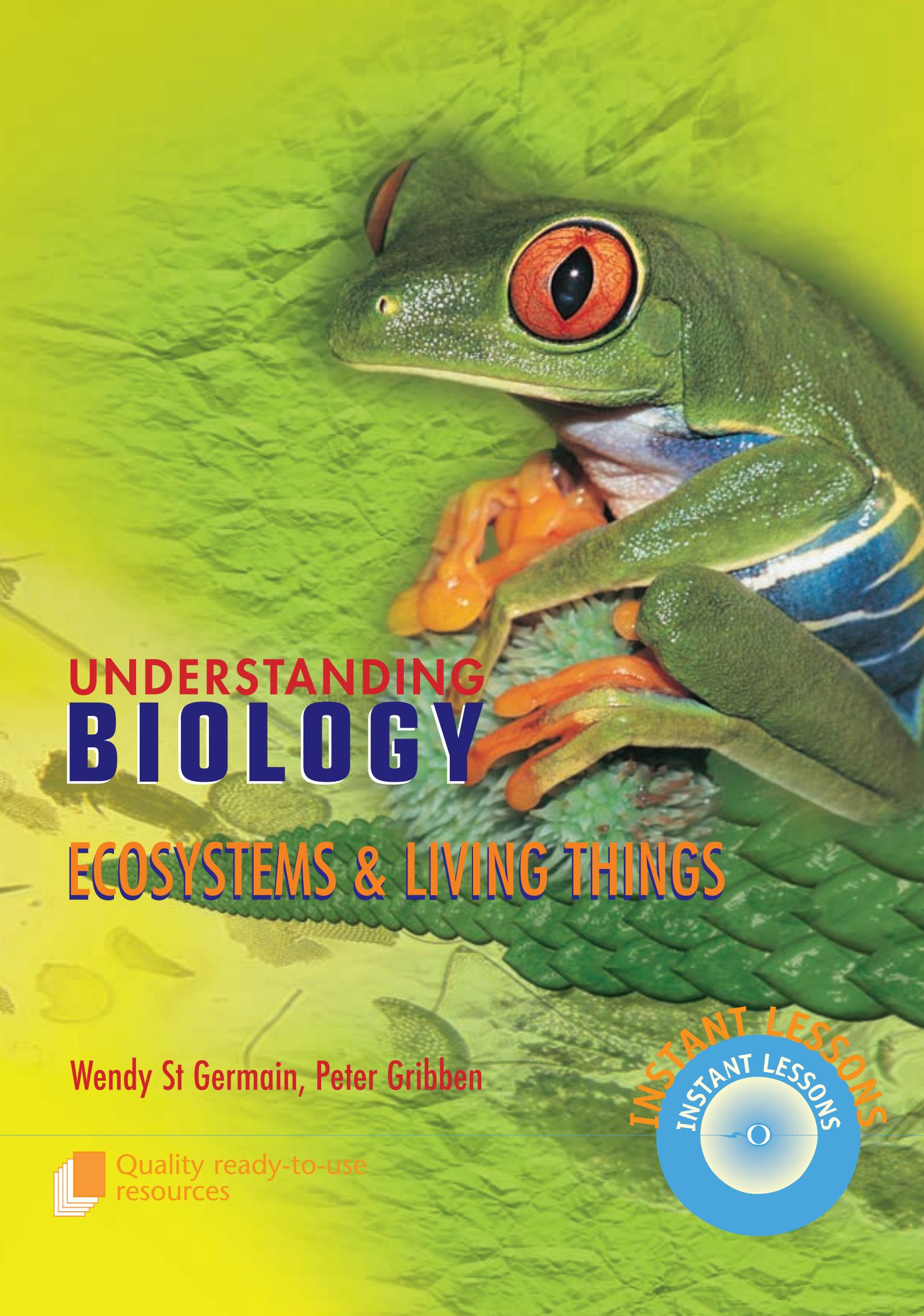




# UNDERSTANDING **BIOLOGY**

## ECOSYSTEMS & LIVING THINGS

Wendy St Germain, Peter Gribben



Quality ready-to-use  
resources





# UNDERSTANDING **BIOLOGY** ECOSYSTEMS & LIVING THINGS

Wendy St Germain, Peter Gribben

This series was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover, Final Page Editing and Design provided page design and typesetting services and David Dickson supplied the illustrations.

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Year 10–12 Biology teachers often report that their students lack basic knowledge of key scientific ideas, which hampers student progress in this subject. The authors of this series have developed activities for topics that, from their experience, they believe are poorly understood by students, as well as for concepts that are particularly crucial to an appreciation of the importance of modern scientific knowledge such as biomaterials, biotechnology and cloning.

In some units, students are required to investigate a topic and provide opinions, and these will make ideal homework or assignment work. Other units are fully self-contained and will test students' understanding of key scientific terms and how this language is used in context.

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- 1 Read the passage and then write each term in *italics* beside its definition in the table below.

Biology is the study of living things. Another name for a living thing is an organism. *Biologists classify* (sort) things according to their form (shape) and structure (how parts are arranged). All living things have characteristics (features) in common. The *characteristics* of all living things are: moving, sensing, growing (and eventually dying), *reproducing*, eating, respiring and *excreting* (discharging) of wastes. All living creatures are made of cells. An organism with one cell is said to be unicellular, whereas those with lots of cells are *multicellular*. There are very small living things that can only be seen using a microscope. These are known as *microbes*.

|   | Definition                                            | Term |
|---|-------------------------------------------------------|------|
| a | Scientists who study living things                    |      |
| b | Distinguishing properties of organisms                |      |
| c | Eliminating unwanted substances from an organism      |      |
| d | Making new offspring                                  |      |
| e | Consists of many cells                                |      |
| f | To arrange things in groups according to similarities |      |
| g | Tiny organisms, unable to be seen by the unaided eye  |      |

- 2 What does a microscope do?\_\_\_\_\_

All living things have features in common, but some living things resemble each other more than others. Because there are many millions of different types of organisms, we put them into groups in order to study them. There are many different ways to sort living things into groups. We could sort them according to where they live, or what they eat, their body coverings (e.g. feathers, fur, skin or scales), how they reproduce or how many legs they have.

- 3 Write one characteristic (not already mentioned) that we could use to classify living things.

Sometimes two organisms look alike, but they may not be related. For example, sharks and dolphins have similarities. If we think more carefully about dolphins, we find they also have some resemblances to humans.

- 4 Write two similarities between dolphins and sharks.

\_\_\_\_\_  
\_\_\_\_\_

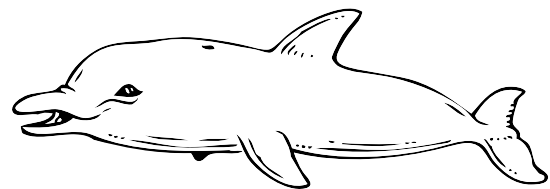


Figure 1

- 5 Write two similarities between dolphins and humans.

\_\_\_\_\_

- 6 Mark the following statements as true or false. Then correct the false statements so that they are true.

- a If two organisms look like each other, they must be related.\_\_\_\_\_
- b It is correct to say 'one fungus' and 'two fungi'.\_\_\_\_\_
- c All animals that live in the sea are classified as fish.\_\_\_\_\_
- d Microbes are large living creatures.\_\_\_\_\_
- e Humans don't resemble any other living creatures.\_\_\_\_\_
- f All living things are made of cells.\_\_\_\_\_
- g All living things have some features in common.\_\_\_\_\_
- h All birds can fly.\_\_\_\_\_
- i A group of dolphins is called a 'pod'.\_\_\_\_\_



Cells would not have been discovered without the development of the microscope. Antonie van Leeuwenhoek (1632–1723) is often credited with the invention of the microscope. Robert Hooke (1635–1703) is the first recorded person to investigate the structure of cells. In 1665 he investigated the structure of a section of cork and described the repeating units he saw as *cells*.

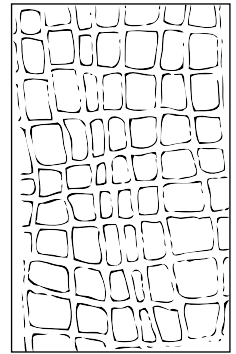


Figure 1

- 1 Why are microscopes essential in the study of cells? \_\_\_\_\_

Hooke stated, ‘... it had a very little solid substance, in comparison of the empty cavity that was contained between. Partitions of these pores were near as thin in their proportion to their pores, as those thin films of wax in a honeycomb.’ Figure 1 is similar to a section of one of his drawings of cork. Hooke didn’t see any of the internal organelles of cells because cork is basically dead wood.

- 2 Define *cavity*. \_\_\_\_\_

- 3 Discuss the comparison of cell structure with a honeycomb. \_\_\_\_\_

- 4 The interior of a cell is made up of tiny, specialised parts called *organelles*. What happened to the internal organelles of the cork cells Hooke studied? Where did they go?

Intense work on cell theory only started in the first half of the 19<sup>th</sup> century. It is likely that the lack of improved microscopes and a reluctance to take on new ideas caused a delay in research into cells. In 1831 Robert Brown (1773–1858), a Scottish botanist, recorded the existence of the nucleus in a range of plant cells. A German botanist, Matthias Schlieden (1804–81), stated in 1838 that plants are composed of cells. Theodor Schwann (1810–82), a German physiologist, advanced a theory in 1839 that animals were also composed of cells. Schlieden and Schwann were not the first people to come up with the concept of cells, but put forward their ideas with clarity and reasoning. In 1858 Rudolf Vichow (1821–1902), a German pathologist, put forward the idea ‘*omnis cellula e cellula*’, that is, all cells arise from cells that already exist.

- 5 Suggest reasons why there was a gap of two centuries between Robert Hooke’s drawing of cells and the general acceptance of cell theory by the mid 19<sup>th</sup> century.

- 6 Write each term beside its correct definition: *biogenesis*, *botany*, *cell*, *organelle*, *nucleus*, *pathology*, *physiology*.

**a** A specialised part of a cell that is comparable to an organ. \_\_\_\_\_

**b** The branch of biology that deals with the study of plants. \_\_\_\_\_

**c** The study of the functioning of organisms. \_\_\_\_\_

**d** Part of cell that controls the functions of a cell. \_\_\_\_\_

**e** Basic unit of living things. \_\_\_\_\_

**f** Study of causes and effects of diseases. \_\_\_\_\_

**g** All life comes from pre-existing life. \_\_\_\_\_

- 7 Describe one contribution to cell theory made by each of these scientists.

**a** Robert Hooke \_\_\_\_\_

**b** Robert Brown \_\_\_\_\_

**c** Matthias Schlieden \_\_\_\_\_

**d** Theodor Schwann \_\_\_\_\_

**e** Rudolf Vichow \_\_\_\_\_

**f** Antonie van Leeuwenhoek \_\_\_\_\_



Most plant and animal cells are less than 0.000 02 m in diameter and are not visible to the unaided eye. It has taken the development of microscopes to enable the close study of cells in order to support cell theory. In scientific method, once a hypothesis is stated (in this case, that all life is made up of cells), it is tested. Scientists have been using sophisticated microscopes to look closely at living things for many years. All of their findings have reinforced the hypothesis of The Cell Theory.

- 1 Define *support*. \_\_\_\_\_
- 2 Define *hypothesis*. \_\_\_\_\_

Nowadays, two main types of microscopes are used. The light microscope uses white (visible) light and glass lenses. The electron microscope uses a beam of electrons focused on a fluorescent screen or photographic plate to produce an image.

- 3 Below are listed the three important attributes of microscopes. Define each one.
  - a magnification \_\_\_\_\_
  - b resolution \_\_\_\_\_
  - c contrast \_\_\_\_\_

By the 1880s, microscopes existed that had magnifications in the range of 500 to 1000 times and very good resolution. Many parts of cells are transparent and are difficult to see.

Scientists developed methods to increase the contrast between parts of cells.

One method used was to add a dye to a cell. Not all parts of the cell take up the dye to the same extent, and some cell structures become coloured more strongly than others.

Staining cells with dye kills the cells, which means that the processes occurring within cells cannot be monitored if the cells are stained. Phase contrast microscopy (introduced in 1934 by the Dutch physicist Frits Zernike) increases the contrast between living cells and their surroundings, so staining can be avoided. Electron microscopes can magnify up to 100 000 times, but cannot be used to study living things because they look at specimens that are in a vacuum.

- 4 Which has the greater magnification, light microscopes or electron microscopes? \_\_\_\_\_
- 5 When examining cells, why might phase microscopy be preferred to staining?  
\_\_\_\_\_  
\_\_\_\_\_
- 6 Outline the differences in how light and electron microscopes work.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
- 7 What is a *vacuum*? \_\_\_\_\_
- 8 Describe why specimens examined under an electron microscope are placed in a vacuum.  
\_\_\_\_\_  
\_\_\_\_\_
- 9 Why are some specimens dyed before examining them under a microscope?  
\_\_\_\_\_  
\_\_\_\_\_
- 10 Although the electron microscope gives far larger and more detailed images than the light microscope, the light microscope is still much more common than the electron microscope. Suggest reasons for this.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

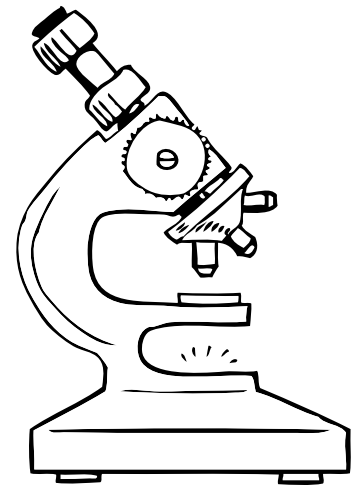


Figure 1



Cells are the basic units of life because they are found in all living things. Cells contain parts which have special functions – these parts are called *organelles*.

Figure 1 shows a simplified version of two types of cells.

- 1 Cell type 'A' represents a:  
\_\_\_\_\_
- 2 Cell type 'B' represents a:  
\_\_\_\_\_
- 3 Identify the organelles shown in figure 1 and described below.

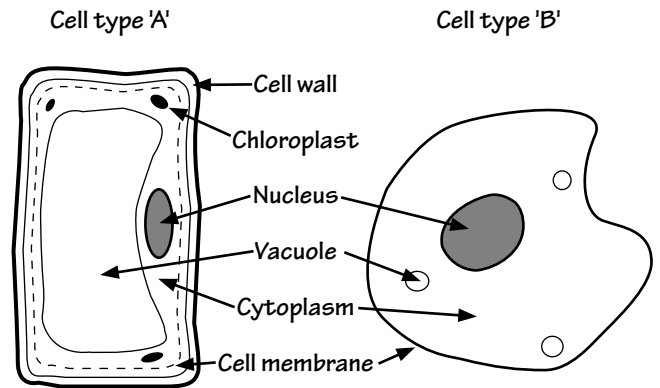


Figure 1

- a Tiny body in a plant cell, containing chlorophyll \_\_\_\_\_
- b Fluid-filled sac in the cytoplasm \_\_\_\_\_
- c Structure made of cellulose, only found in plant cells \_\_\_\_\_
- d Control centre of the cell \_\_\_\_\_
- e Liquid area between the cell membrane and the nucleus \_\_\_\_\_
- f Outer, permeable skin of a cell \_\_\_\_\_

There are many cell organelles that cannot be viewed with light microscopes. Electron micrographs must be studied to see their details.

- 4 Write each of these organelles beside its correct description:  
*endoplasmic reticulum, Golgi bodies, lysosomes, mitochondria, nucleolus, ribosomes.*
  - a Modifies and distributes proteins (and other substances) made in the cell \_\_\_\_\_
  - b Gives large surface area for reactions and fluid storage \_\_\_\_\_
  - c Small particles lining the surface of the endoplasmic reticulum \_\_\_\_\_
  - d Round sacs that contain digestive enzymes \_\_\_\_\_
  - e 'Powerhouses' that produce energy by cellular respiration \_\_\_\_\_
  - f Oval/round body found in the nucleus \_\_\_\_\_
- 5 Discuss this statement: *A cell can resemble the human body, in that it is composed of smaller units that act as a nervous system, skin, digestive system and skeleton.*  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- 6 Is it possible for a single cell to exist as a living entity? Or must an organism be made of many cells? Explain your answer.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_



All organisms are composed of microscopic units – cells. The major constituent of most cells is water,  $H_2O$ . Many cells are more than two-thirds water. Water, because it is such a good solvent, dissolves many chemicals and allows them to react and move in and out of cells.

Carbon dioxide ( $CO_2$ ) is a gas formed during respiration. It readily dissolves in water and also comes out of solution easily. Carbon dioxide and water are the raw materials used by green plants to make complex compounds. Oxygen ( $O_2$ ) is needed by most organisms. It is not as soluble in water as carbon dioxide is.

**1** Give the formula for a molecule of these substances.

**a** Water \_\_\_\_\_

**b** Carbon dioxide \_\_\_\_\_

**c** Oxygen \_\_\_\_\_

Small amounts of minerals (salts) are needed for chemical processes in cells. These minerals include sodium ( $Na^+$ ), potassium ( $K^+$ ) and calcium ( $Ca^{2+}$ ) ions. An ion is an electrically charged particle – that is, a charged atom or group of atoms. The attraction between different ions results in chemical reactions and new substances are formed. With the exception of the previously mentioned examples, most of the molecules present in cells are based on the element carbon (C), and are called organic compounds.

Glucose,  $C_6H_{12}O_6$ , is a simple sugar or monosaccharide. It plays an important role in energy transformations in organisms.

**2** Which elements are present in glucose? \_\_\_\_\_

**3** What causes ions to form new substances? \_\_\_\_\_

Glucose is a carbohydrate; it contains carbon, hydrogen and oxygen. Hydrogen and oxygen are in the same ratio as in water, that is, two hydrogen atoms to one oxygen atom.

Respiration describes the chemical processes involved when a cell breaks down a food source in order to transfer the energy it contains. This energy is used to fuel cellular processes. When a carbohydrate is respired aerobically (with oxygen) by cells, it gives off energy, carbon dioxide and water. Glucose molecules can be joined together to make more complex sugars such as disaccharides (sugar made up of two units, *di* meaning ‘two’) or polysaccharides (sugar made up of many units, *poly* meaning ‘many’). Glucose can act as a store of energy. In plants, starch is formed from thousands of glucose units. These large starch molecules are types of carbohydrates.

**4** Does forming starch from glucose molecules require the input of energy or the release of energy? Explain.

Like plants, animals can store excess glucose in the form of starches. However, animal starch is not exactly the same as plant starch so it is given a different name, glycogen. Glycogen is also formed from glucose molecules.

Both starch and glycogen can be regarded as polymers – large molecules made up of similar, repeating molecules. When an organism needs energy, it can break down the complex molecule into glucose units and release energy during this process, as shown in figure 1.

**5** Which element is the basis of ‘organic’ compounds?

\_\_\_\_\_

**6** Write the names of three ions found in cells.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**7** Write a definition for *aerobic*. \_\_\_\_\_

\_\_\_\_\_

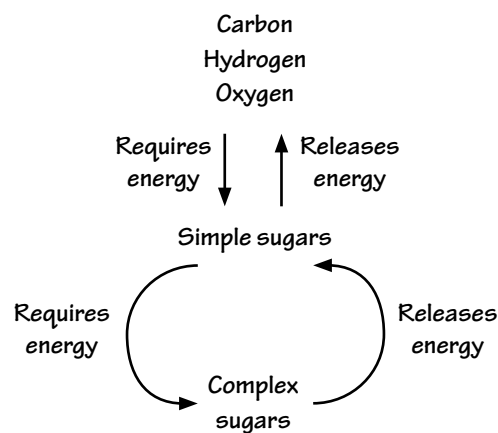
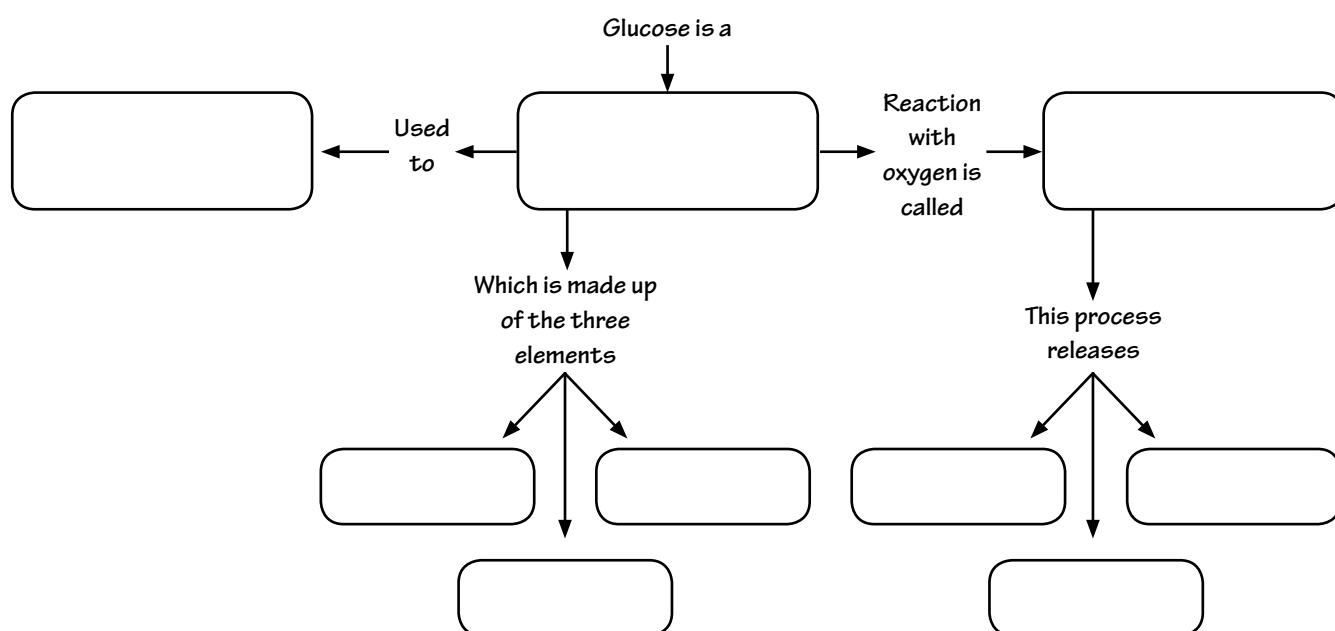


Figure 1



- 8** Outline the differences between a monosaccharide, a disaccharide and a polysaccharide.
- \_\_\_\_\_
- \_\_\_\_\_
- 9** Which type of substance – monosaccharide, disaccharide or polysaccharide – would be the easiest to break down for energy? Explain your answer
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- 10** Write a definition for *carbohydrate*. \_\_\_\_\_
- \_\_\_\_\_
- 11** Name three foods that are rich in carbohydrates. \_\_\_\_\_
- \_\_\_\_\_
- 12** Athletes often eat foods rich in carbohydrates before a sporting event. Suggest a reason for this.
- \_\_\_\_\_
- 13** As well as carbohydrates, athletes also eat protein-rich foods such as meat. Suggest a reason for this.
- \_\_\_\_\_
- 14** Write an example of an inorganic substance. State why it is considered inorganic.
- \_\_\_\_\_
- \_\_\_\_\_
- 15** Write the chemical formula for glucose. \_\_\_\_\_
- 16** Complete the chart below, using these terms: *carbohydrate, carbon, carbon dioxide, energy, hydrogen, oxygen, respiration, store food, water*.





Biologists sometimes group organisms according to whether they can maintain a steady body temperature independent of external temperatures, or whether their body temperature varies with the external temperature. Endotherms (previously known as ‘warm-blooded’ or ‘homeotherms’) are animals such as birds and mammals that get their (steady) body warmth from their cell metabolism. (*Endo* means ‘internal’.) Ectotherms (‘cold-blooded’ or ‘poikilotherms’) include animals such as reptiles and fish that control their body heat by mainly external means. (*Ecto* means ‘outside’.)

A researcher wanted to find out whether a certain type of organism changed its body temperature in response to changes in air temperature. Her results are shown in the table below.

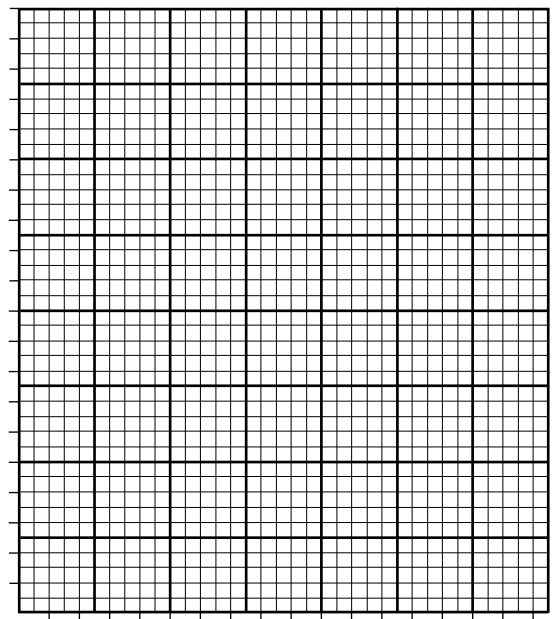
| Air temp. °C           | -2 | 0 | 2 | 4 | 6 | 8  | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28 |
|------------------------|----|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
| Organism body temp. °C | 2  | 4 | 5 | 7 | 9 | 12 | 14 | 17 | 19 | 21 | 23 | 25 | 28 | 31 | 35 | 37 |

- 1 Use the graph paper at the right and the information in the table above to draw a ‘best fit’ line graph. Plot air temperature on the X-axis and body temperature on the Y-axis. Label the axes.
- 2 Is the animal studied an endotherm or ectotherm? Justify your answer.  

---

---
- 3 Predict the organism’s body temperature at an air temperature of 27°C. Show on your graph how you obtained your answer.  

---



Because ectotherms do not use the energy generated by their metabolisms to regulate their body temperatures, they only need to eat 10% to 30% of the food required by an endotherm of equivalent size. Many ectotherms use the heat of the Sun to raise their body temperatures at the beginning of the day.

Although they need less food to maintain themselves, ectotherms are active for less of the day than endotherms. Ectotherms can survive in areas where there are not enough resources for endotherms to thrive. They can also reproduce much more quickly than endotherms, because they can accumulate the energy needed very quickly.

- 4 What is meant by *accumulate*? \_\_\_\_\_
- 5 What is meant by *thrive*? \_\_\_\_\_
- 6 Name an organism (ectotherm) that uses the Sun to warm itself. \_\_\_\_\_
- 7 Explain why one particular environment could support 3–10 times the number of ectotherms as endotherms of similar size.  

---
- 8 Even though resources may be scarce in a particular habitat, sometimes endotherms (especially predators) can dominate over ectotherms. Suggest a possible reason for this.  

---

Ambient (environmental) temperatures are sometime warmer or colder than is comfortable for a particular organism. Endothermic organisms have many ways to adjust their systems to either shed heat or to conserve it.

- 9 Study the following examples and state whether heat is being lost or saved, and how it works.
  - a A dog panting \_\_\_\_\_
  - b In cold weather, blood vessels near the skin constrict (get smaller). \_\_\_\_\_
  - c A desert animal making a burrow \_\_\_\_\_



As we can see from the millions of different lifeforms on Earth, organisms have adapted to survive in their particular habitats.

**Adaptation:** biological process (brought about by natural selection) by which an animal or plant becomes fitted to its environment.

- 1 Outline what is meant by *natural selection* and how it relates to adaptation.

---

---

---

These adaptations can be in an animal's behaviour, structure or physiology (bodily processes). For example, migration to a different location to find food or better weather can help a species survive.

- 2 Give one other reason for an animal to migrate. \_\_\_\_\_

- 3 Name a species that migrates and explain its reasons for doing this. \_\_\_\_\_

---

---

- 4 *The study of the function of the constituent parts of organisms* – what does this define?

---

The owner of some grazing land seeded it with a mixture of grasses and legumes. The pasture was then divided into two parts. One part was heavily grazed by cattle and the other part was kept free of livestock and used to produce hay.

After three years, the owner took samples of the grasses and legumes from both parts and re-planted them in an experimental garden, ensuring that all samples were exposed to the same environmental conditions. The specimens that were taken from the heavily grazed part showed stunted and rambling growth, while those from the ungrazed part grew larger and more upright.

- 5 Explain why the differentiation in plants occurred, keeping in mind the fact that the same batch of seed was used for both parts of the pasture.

---

---

Consider the bodily adaptations of marsupials. Kangaroos are *terrestrial* grazers and their hind limbs are larger and more powerful than their forelimbs. The forelimbs of *arboreal* marsupials (and also those that burrow) are often nearly as large as their hind limbs. Kangaroos have large tails, used mainly to help them balance. In tree-dwelling species, the tail may also be *prehensile*.

- 6 Define these terms.

**a** terrestrial \_\_\_\_\_

**b** arboreal \_\_\_\_\_

**c** prehensile \_\_\_\_\_

Consider plants that rely on insects to pollinate them. They often have brightly coloured flowers.

- 7 Describe one other adaptation present in some plants that attract insects to aid pollination.

---

---



## Cryptic colouration

Many animals are difficult to distinguish from their backgrounds. This is an adaptive characteristic that helps the organisms to escape predators. Scorpion fish live on coral or on rocky sea bottoms, and use a combination of cryptic colouration and shape to help them survive.

- 8** What is another name for cryptic colouration? \_\_\_\_\_
- 9** Give another example of an animal that uses cryptic colouration, and describe how this helps the animal's survival.

---

---

Some animals, especially insects, have evolved patterns and colourations that make them stand out from their surroundings. These animals often have some characteristic that is unpleasant to predators; they may contain poison, taste bad or sting. Predators usually learn after a few disagreeable experiences to leave these animals alone. Some species that don't have protective characteristics may mimic (imitate) the appearance of an unpalatable species.

- 10** Why might it be an advantage for a harmless organism to mimic the appearance of a species that does have 'warning' colourations?

---

- 11** Imagine that a particular type of herbivore faces a lack of vegetation to eat in its current habitat, although there are many smaller animals present. In addition to changing its behaviour, what other adaptations would be needed before this species could be a successful carnivore?

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For a population that contains variations, the selection pressure causes evolutionary change. We cannot always be correct when we infer that particular adaptations assist an organism to live in a particular habitat. Consider dinosaurs – scientists are still undecided as to whether they were *endotherms* (body temperature regulated internally, 'warm-blooded') or *ectotherms* (body temperature dependent on external factors, 'cold-blooded').

In humans, sickle-cell anaemia is a disease that can cause death. However, in areas where malaria is present this disease can be an advantage as it confers some resistance to malaria.

- 12** Consider all of the breeds of dog that exist. Outline the characteristics of one particular breed and suggest possible reasons for these adaptations.

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An adaptation in an organism can be described as a change that helps this organism to survive in a particular environment.

Adaptations can be in the organism's:

- structure (e.g. body shape)
- physiology (e.g. internal chemistry)
- behaviour (e.g. what the organism does).

Australia has a variety of ecosystems.

Dry environments can include:

- grassland
- scrub
- heath.

- 13** Name one other type of dry environment. \_\_\_\_\_

Where there are greater amounts of rainfall, forests and woodlands can exist. There are a large number of plants and animals unique to Australia that have adapted to the special conditions that exist.



- 14** These plant adaptations are found in dry areas. Explain how each adaptation helps with survival.
- a** spines on stems \_\_\_\_\_
  - b** deep root systems \_\_\_\_\_
  - c** small leaves \_\_\_\_\_
- 15** Adaptations can be *structural*, *physiological* or *behavioural*. Which type of adaptation is present in these examples?
- a** spines on stems \_\_\_\_\_
  - b** deep root systems \_\_\_\_\_
  - c** small leaves \_\_\_\_\_
- 16 a** Identify two other types of adaptations in plants, not already mentioned.
- \_\_\_\_\_
- b** Outline how these adaptations help the plants survive.
- \_\_\_\_\_
- \_\_\_\_\_
- c** State the type of adaptation present. \_\_\_\_\_

Perhaps one of the harshest environments is the desert, where water is scarce and temperatures are high during the day. Kangaroos conserve energy and moisture during the day by staying in the shade and coming out to feed when it is cooler.

- 17** Suggest reasons why some small marsupials live in burrows during the day and come out to feed at night.
- \_\_\_\_\_
- \_\_\_\_\_
- 18** Identify and classify the adaptations shown in the following example.
- Desert animals ensure that they make the best use of the little water they get. They sweat very little, and have very efficient kidneys to reduce the amount of urine excreted. Many also have a metabolism that can form water from the relatively dry plants they consume.*
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- 19** Many fish are *streamlined* to enable them to swim faster. What does 'streamlined' mean? Why is streamlining important for fish?
- \_\_\_\_\_
- \_\_\_\_\_
- 20** Kangaroos are *crepuscular*. Find out what this means.
- \_\_\_\_\_
- 21 a** Identify two other types of adaptations in animals, not already mentioned.
- \_\_\_\_\_
- b** Outline how these adaptations help the animals survive.
- \_\_\_\_\_
- \_\_\_\_\_
- c** State the type of adaptation present. \_\_\_\_\_

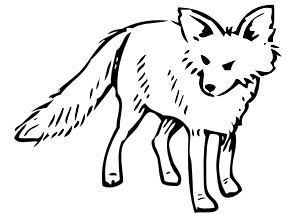


When there are limited resources in a particular ecosystem, the organisms present will compete for them. These resources can include food, shelter, space and water. If one species is more successful (or better suited) to the local environment, it will succeed at the expense of the less successful species.

- 1 What might happen to the less successful species over time?
- 

If both species are approximately equally successful, they will coexist and their numbers will remain relatively stable.

Introduced species can cause problems. They often have no competitors or are better suited to a local ecosystem than the indigenous species are. Foxes belong to the genus *Vulpes* and are not native to Australia. They were introduced from England by early settlers. Foxes have displaced native carnivores in many parts of Australia.



- 2 Outline why foxes have become a problem in Australia.
- 

Rabbits (family *Leporidae*) are another introduced species. Even at low population levels they can affect the types of vegetation growing and cause soil erosion. In countries such as England and Canada, foxes and rabbits seldom create these problems.



- 3 What other problems are caused by rabbits?
- 

- 4 Suggest why animals like foxes and rabbits do not cause major environmental problems in the countries to which they are native.
- 

- 5 Read the following passage and write a definition for each word in *italics*.

Rabbits are *prolific* breeders. They become *sexually mature* at around six months of age. Under ideal conditions, female rabbits can have four or more *litters* a year. Their *gestation period* is around four weeks.

- a prolific \_\_\_\_\_
- b sexually mature \_\_\_\_\_
- c litter \_\_\_\_\_
- d gestation period \_\_\_\_\_

Cattle and sheep have hard hooves that are not adapted for the Australian environment. Unlike kangaroos, they cause a lot of damage to soil.

- 6 Outline how cattle and sheep can damage soil. \_\_\_\_\_
- 

If there is pressure on an organism from its environment, over an extended period of time that organism may evolve and change to become better suited to its environment. Competition from other species can be one factor that influences evolution.

Consider the following scenario. An area has around 5000 herbivores, but there is sufficient vegetation for around 7000 herbivores. A small but stable population of carnivores preys on the herbivores. Hunting by humans over a short period reduces the number of carnivores dramatically.

- 7 Discuss the possible results of this human intervention.
- 

- 8 Outline the resources for which plants might compete in this area.
- 

- 9 *A community of organisms, their environment and interactions* – what does this define?
-



In nature, animals breed, die of natural causes and are also eaten by predators. If, in a group of animals, the number dying is greater than the number being born, then this group will eventually die out.

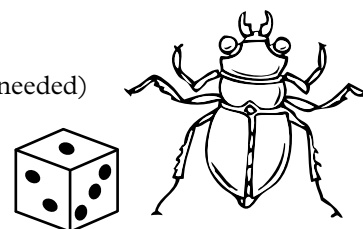
- 1 What would happen to this group of animals if the number being born was greater than the number dying?

## Beetle simulation

This exercise is a simulation to investigate how one factor (colour) influences the make-up of a group of 'beetles'. The beetles are all the same species (i.e. can mate and produce fertile offspring) but can be of different colours – blue, yellow or green. The colours of offspring occur according to a set ratio.

For this simulation you need:

- 2 sets of 'beetle' cards (10 blue, 10 yellow, 10 green and 30 to be coloured as needed)
- a six-sided die.



## Your task

Establish a beetle population and simulate cycles of reproduction and population.

## Assumptions about beetle population:

- i There are 30 beetles in the original population – 10 blue, 10 yellow and 10 green.
- ii Each year the beetles mate randomly. Each pair produces one offspring; that is, a total of 15 new beetles are born during one year.
- iii The colour of the offspring is determined according to table 1 (below).
- iv During one year, a predator kills one-third of the total population (15, as this includes the year's offspring). On average, its prey consists of beetles in the ratio of 3 blue : 2 green : 1 yellow. This means that the population remains steady at 30 at the start of each breeding season.

Table 1

| Colour of parents | Colour of offspring             | Colour selection of offspring – roll of die                  |
|-------------------|---------------------------------|--------------------------------------------------------------|
| Blue x blue       | All blue                        | n/a                                                          |
| Yellow x yellow   | All yellow                      | n/a                                                          |
| Blue x yellow     | All green                       | n/a                                                          |
| Green x yellow    | 50% green, 50% yellow           | 1, 2, 3 = green<br>4, 5, 6 = yellow                          |
| Green x blue      | 50% green, 50% blue             | 1, 2, 3 = green<br>4, 5, 6 = blue                            |
| Green x green     | 50% green, 25% yellow, 25% blue | 1, 2, 3 = green<br>4 = yellow<br>5 = blue<br>6 = throw again |

## Instructions

- 2 Start your beetle population by selecting 30 beetle cards and colouring 10 blue, 10 yellow and 10 green.
- 3 Keep the remaining 30 cards as a source of beetle offspring.
- 4 To represent *random* mating, shuffle your 30 cards thoroughly. Deal them in 15 pairs. (We are assuming that each beetle can act as either male or female.)
- 5 For each resulting pair, add another card representing the offspring. Use table 1 to decide the appropriate colour. If there is a choice of colours, throw the die and refer to the third column of table 1 to decide the colour.



- 6** To simulate predation, you now need to take away 15 beetles. Roll the die, and remove a beetle of the colour corresponding to the number on the die. Use table 2 below to decide which beetles to remove.

**Table 2**

| Number on die | Remove beetle of this colour |
|---------------|------------------------------|
| 1, 2, 3       | Blue                         |
| 4, 5          | Green                        |
| 6             | Yellow                       |

- 7** Repeat this until 15 beetles have been removed. Record the resulting population in table form in the space below.
- 8** Repeat the whole procedure for a total of 10 years.

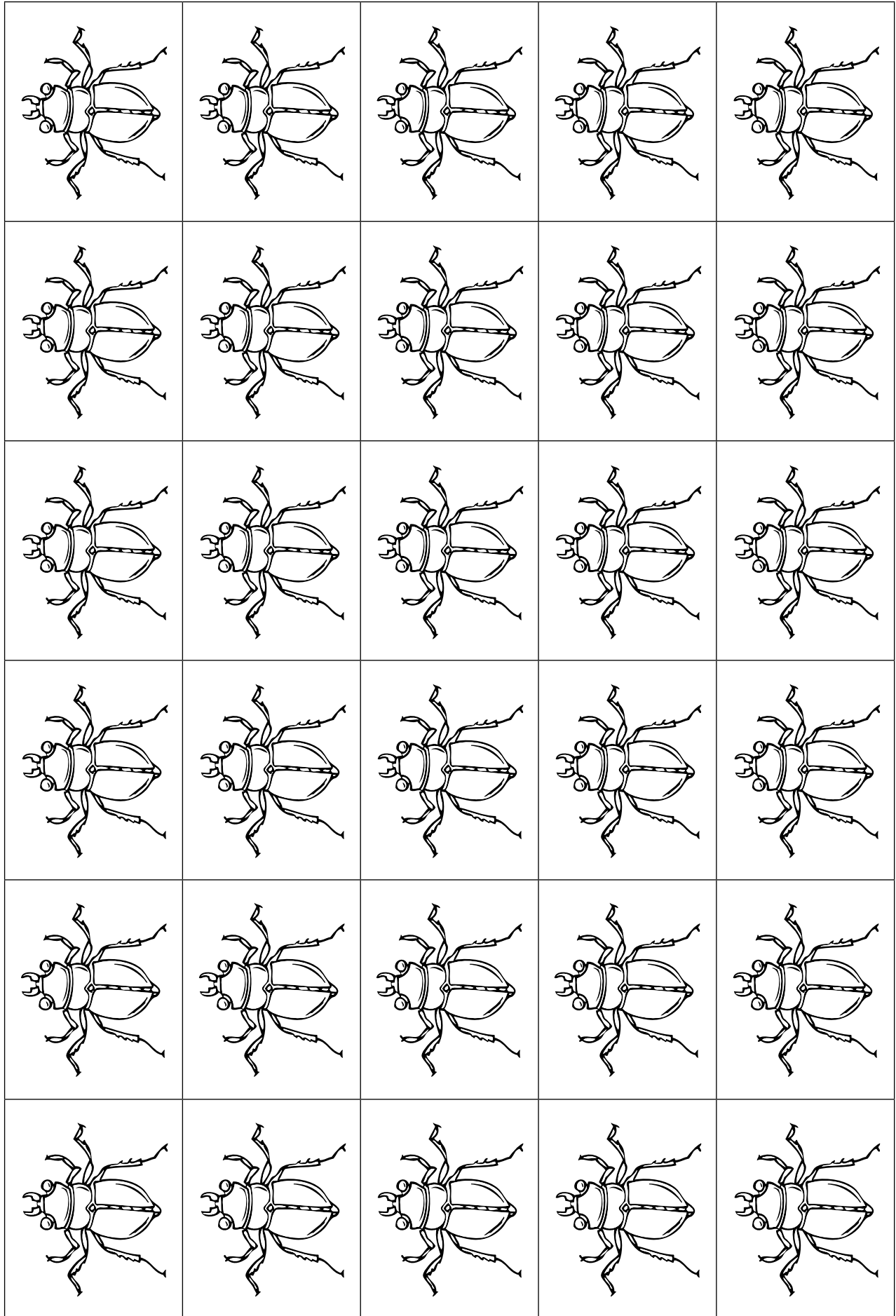
**9** Comment on your results. \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**10** Why must we begin this simulation by making certain assumptions about the population?  
\_\_\_\_\_

**11** What does *random* mean? \_\_\_\_\_

**12** This simulation makes certain assumptions. Identify two assumptions that are made regarding the changes in the beetle population.  
\_\_\_\_\_  
\_\_\_\_\_

**13** Suggest ways in which this sort of simulated 'modelling' might be useful to scientists.  
\_\_\_\_\_





Organisms have changed over time. Most people think these changes are a result of the environment acting on these organisms. This is not a new idea – it was first proposed over 2000 years ago that all organisms evolved from other organisms. The study of fossils has been a major factor in the theories of evolution that have been put forward since then. Another supporting fact is that living things share similar characteristics and similar chemistry.

- 1** List five things that organisms have in common.

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The chemistry of living things has been described as ‘CHON-based’. This term refers to the fact that C, H, O and N are the elements found most commonly in organisms.

- 2** Write the names of the elements represented by these symbols.

**a** C \_\_\_\_\_ **b** H \_\_\_\_\_ **c** O \_\_\_\_\_ **d** N \_\_\_\_\_

It has been theorised that life began on Earth in the sea around 4 billion years ago. The first living things were algae and bacteria, which are thought to have been the first organisms responsible for photosynthesis.

- 3** Outline the process of photosynthesis. \_\_\_\_\_

---

- 4** ‘Bacteria’ and ‘larvae’ are plural; that is, refer to more than one organism.

Write the singular forms of bacteria and larvae. \_\_\_\_\_

- 5** Traces of algae identical to those present today have been found from over 3 billion years ago. True or false?

- 6** What are *marine* organisms? \_\_\_\_\_

Around 600–700 million years ago, more complex *invertebrates* began to appear. Some of these invertebrates developed an *exoskeleton*.

- 7** Define *invertebrate*. \_\_\_\_\_

- 8** Define: *exoskeleton*. \_\_\_\_\_

- 9** Give one example of an invertebrate and one example of an animal with an exoskeleton.

---

Around 500 million years ago, vertebrates (chordates) began to appear in the sea. About 100 million years later, plants began to appear on land and vertebrates first started to move from the sea to land. Amphibians first appeared in this era. These are animals that live on land, lay their eggs in water and their young live in water before changing and then living on land.

- 10** List two amphibians. \_\_\_\_\_

Reptiles began to breed on land approximately 300 million years ago. Around the same time, insects developed flight.

- 11** List three physical properties of reptiles. \_\_\_\_\_

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- 12** Name three reptiles. \_\_\_\_\_

- 13** Name three insects that can fly. \_\_\_\_\_

- 14** Mark the following statements as true or false. Then correct the false statements so that they are true.

**a** Only some living things are made up of cells. \_\_\_\_\_

**b** All living things use DNA/RNA as the basis for inheritance of characteristics. \_\_\_\_\_

**c** Evolution can be described as a gradual change and adaptation of an organism over time. \_\_\_\_\_

- 15** Over time, do all species eventually become extinct? Explain your answer.

---

- 16** Who is known as ‘The Father of Evolution’? \_\_\_\_\_



Plants and animals usually only change their characteristics in response to a change in their environment.

- 1** Explain what is meant by the *characteristics* of an organism. Include an example.

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Many animals, such as sharks and crocodiles, have remained relatively unchanged over millions of years because they fit their specialised ecological niche so well. A similar example in plants is the Wollemi pine. These organisms will not change unless they are forced to do so.

- 2** List three features of a crocodile that help it to survive in its environment.

---

- 3** *The process by which different kinds of living organisms are believed to have developed, especially by natural selection* – what is defined by this statement?

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Plants need water (H<sub>2</sub>O), carbon dioxide (CO<sub>2</sub>), sunlight and some minerals to grow.

- 4** Which part of a plant absorbs water and minerals? \_\_\_\_\_

- 5** Which parts of a plant absorb solar energy and carbon dioxide? \_\_\_\_\_

Minerals (sometimes called *ions*) are taken by the plant from the soil. Typical soils in Australia are low in phosphorus (P), so native plants have developed to exist on small amounts of phosphorus. However, introduced species of plants thrive where there are high concentrations of phosphorus.

Some modern detergents have been specially formulated to have little or no phosphorus. However, most modern detergents contain high levels of phosphorus. If this phosphorus is allowed into the environment, introduced plants will have an advantage, and eventually displace native plants.

- 6** Explain what is meant by the following terms, as used in the paragraphs above.

**a** native \_\_\_\_\_

**b** solar energy \_\_\_\_\_

**c** introduced \_\_\_\_\_

**d** detergent \_\_\_\_\_

**e** displace \_\_\_\_\_

**f** thrive \_\_\_\_\_

Common salt is sodium chloride (NaCl). There are many other similar compounds that are called *salts*. The level of salt in soil in many areas of Australia has increased due to human activities. Some plants are not tolerant of elevated salt levels. Other plants, such as saltbush, can cope with these conditions.

- 7** Explain what is meant by the following terms.

**a** tolerant \_\_\_\_\_

**b** elevated \_\_\_\_\_

**c** cope \_\_\_\_\_

- 8** Which chemical compound has the formula H<sub>2</sub>O? \_\_\_\_\_

- 9** Suggest how the presence of a plant in an ecosystem might direct the evolution of an animal.

---

- 10** Suggest how the presence of an animal in an ecosystem might direct the evolution of a plant.

---



Organisms need many things to survive and reproduce. The things that are used by organisms are called resources. The resources for a plant might include sunlight, water, soil and minerals in the soil. Animals need plants for food, water to drink, and oxygen to breathe.

- 1** Name one other resource needed by plants. \_\_\_\_\_
- 2** Name one other resource needed by animals. \_\_\_\_\_

If resources are plentiful, organisms in a particular environment will thrive. It is only when there is a shortage of resources that there will be pressure on a particular organism for survival. Below is a simplified version of Charles Darwin's Theory of Evolution, caused by natural selection.

- Not all organisms in a population are identical.
- Organisms that survive for long enough, reproduce and pass on characteristics to their offspring.
- Some characteristics are more useful in a particular environment.
- The organisms that are more suited to a particular environment are more likely to survive.
- Over a large number of generations, organisms may evolve to suit their environment.

- 3** Below is a set of proposals (not in order of occurrence) about how giraffes evolved. Select the statements that are relevant to the evolution of long necks in giraffes. Write them in a likely sequence; that is, in order of occurrence, from first to most recent.

- a** Giraffes live mainly in African bush country.
- b** Giraffes tend to eat acacia leaves.
- c** The ancestors of giraffes had short necks
- d** Modern day giraffes have long necks.
- e** The supply of acacia leaves in Africa can fall to very low levels.
- f** Some giraffes in the past had longer necks than others.
- g** Leaves on lower branches of acacia trees were easier to reach for the ancestors of giraffes.
- h** Giraffes with longer necks survived to reproduce.
- i** Leaves on lower branches were eaten first, leaving only hard-to-reach leaves on higher branches.
- j** When food supplies were low, giraffes with longer necks could reach food on higher branches.
- k** Giraffes with shorter necks could not reach food on higher branches and died out.
- l** Giraffes can reach over 5 metres in height.
- m** Giraffes have well developed senses.

- 4** Like many other mammals, giraffes have only seven vertebrae in their necks. Suggest how their necks could have grown longer.

Rainforests contain tall trees with broad leaves. The trees need large amounts of rainfall and plentiful sunshine to prosper. Rainforests are very dense areas of vegetation, and are areas of high humidity. Rainfall gradually trickles down from the upper canopy (layer of leaves) to the forest floor. The roots of plants tend to be shallow.

- 5** Explain why plants in a rainforest have to compete for sunlight. \_\_\_\_\_
- 6** Describe two adaptations that would help rainforest plants to access the maximum amount of sunlight. \_\_\_\_\_
- 7** *A material, help or capacity supplied by nature* – what does this define? \_\_\_\_\_
- 8** List the major resources needed by rainforests to survive. \_\_\_\_\_



The belief that living things have changed over time existed well before Charles Darwin (1809–1882) publicly stated it. For many years, people have wondered how fossils are related to current living things. A fossil is what is preserved in the Earth's crust after an organism has died.

- 1** In your own words, describe two types of fossils.

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***Palaeontology:** a science dealing with the life of past geological periods, as known from fossil remains.*

- 2** Why might a scientist studying the evolution of a particular organism be interested in palaeontology?

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Most fossils are found in sedimentary rock. Interpreting the sequence of fossils in layers and subsequently dating them relies on the geological 'Law of superposition'. This means that fossils in lower levels are older than fossils in levels nearer the surface. The sequence of fossils in layers is known as the fossil record.

- 3** What is sedimentary rock? Give an example.

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- 4** Before palaeontologists can draw definite conclusions about a fossil record in one area, they must be sure that the rocks have been undisturbed.

**a** Why? \_\_\_\_\_

**b** What could disturb them? \_\_\_\_\_

**c** How could scientists find out if the area had been disturbed? \_\_\_\_\_

- 5** Palaeontologists have needed to compare layers of sedimentary rock from different parts of the world to work out the sequence of life on Earth. Why is this?

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- 6** Comment on the observation that, in general, the fossils found in older rocks are simpler than those found in newer rocks.

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- 7** The fossil record is not complete and there are gaps in the records of the evolution of many organisms. Suggest reasons why there are gaps.

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- 8** Write a definition for *fossil record*.

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The evolution of the modern horse has been studied using fossil evidence dating back almost 60 million years. The earliest ancestor of the modern horse is referred to as *Hyracotherium* but is more often referred to as *eohippus*, the 'dawn horse'. Some of these evolutionary changes are listed below.

- Increase in overall size
- Teeth becoming larger and more suited to grazing
- Reduction in the number of hooves
- Legs becoming longer
- Muzzle becoming longer
- Brain becoming larger and more complex

**9** Choose one of the above features and discuss how it may have been an advantage for the horse.

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## Transition fossils

Often scientists find that there are 'gaps' in the fossil record of a particular modern organism. This is mainly because fossilisation occurs only rarely, and fossils, once formed, are easily destroyed by the forces of nature. Sometimes fossils are found that seem to have characteristics of two different types of organisms. Fossils of the archaeopteryx have been used to support the argument that birds evolved from dinosaurs. The archaeopteryx (figure 1) had characteristics of both a bird (feathers, a beak and claws) and a dinosaur (teeth and a long, bony tail).

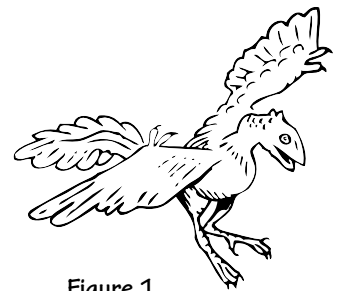


Figure 1

**10** Consider the statement: *The archaeopteryx was not a bird, just a dinosaur with feathers.* Write a response to this statement, based on what you know about transition fossils.

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Lobe-fin fish (order *Crossopterygii*) were fish that lived 400 to 360 million years ago. (Figure 2 shows an example of *Latimeria*, a modern lobe-finned fish.) These fish had lungs and paired, rounded fins. Their skulls had characteristics of amphibians and their vertebral column was midway between that of a fish and an amphibian. They were probably able to pull themselves onto sand bars and mud flats. Labrynthodonts are primitive amphibians that lived about 350 to 210 million years ago. They had many similarities to lobe-fin fish.

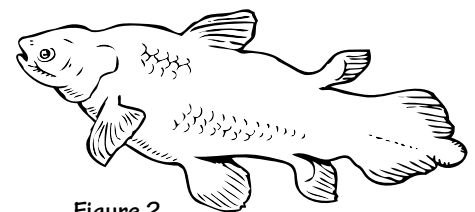


Figure 2

**11** *From the Greek, meaning 'having a double life', they are ectothermic vertebrates that have gilled aquatic larvae and air-breathing adults* – what does this define?

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**12** Explain what is meant by *transition fossils*.

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**13** Describe the physical characteristics that would enable *Crossopterygii* to spend some time on land.

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**14** What do you think motivated *Crossopterygii* to spend time on land?

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Charles Darwin (1809–1882) and Alfred Russell Wallace (1823–1913) independently developed similar ideas about evolution being driven by natural selection. The three major ideas below are fundamental to current theories of evolution.

- Characteristics of organisms change over time.
- Changes in organisms are driven by natural selection.
- Individuals of a species can't evolve, but populations do.

**1** Explain the difference between an *individual* and a *population* of a species. \_\_\_\_\_

**2** What is more important, the survival of an individual or the survival of a population? Explain your answer. \_\_\_\_\_

Populations of developed organisms are made up of many individuals that share many important features. However, all individuals have their own unique differences.

**3** For humans, state three:

**a** similarities. \_\_\_\_\_

**b** differences. \_\_\_\_\_

Darwin's *On the Origin of Species* was published in 1859. Although some of Darwin's ideas on evolution have been replaced, his major proposals are still a good basis on which to form the theory of evolution. These are listed below.

- There are many different physical characteristics (phenotypes) present in a population.
- Reproduction causes the numbers of that population to increase.
- Eventually, the supplies of resources will not support the increasing numbers of a population.
- Those individuals who are better suited to obtain resources will survive.
- The inherited characteristics of the successful organisms will be passed on to their offspring.
- Different environments will be helpful to different types of organisms.

**4** Identify three possible resources that are in demand by organisms in a particular ecosystem. \_\_\_\_\_

Darwin and Wallace were among the first people to notice the effect of isolation on species. On his voyage on *HMS Beagle* (1831–36), Darwin noticed that although the finches in the Galapagos Islands were very similar, they had beaks of different shapes and sizes depending on the specific island on which they were found.

**5** Discuss the possible effects that different types of beaks would have on individual species of finches. \_\_\_\_\_

**6** How might this beak evolution affect and be affected by mating? \_\_\_\_\_

Darwin wondered why there was so much diversity between finches (and many other animals) on a few small islands, 1000 km west of the coast of South America. Many of these species were not found on the mainland. He came to believe that these differences came about, in some part, due to the isolation of these islands from other populated areas.

**7** What is meant by *geographical isolation*? \_\_\_\_\_

**8** Darwin stated, 'Species are not immutable.' Explain what this means in your own words. \_\_\_\_\_



- 9 Name another place that is geographically isolated. How has this isolation influenced the way that its native species have developed?

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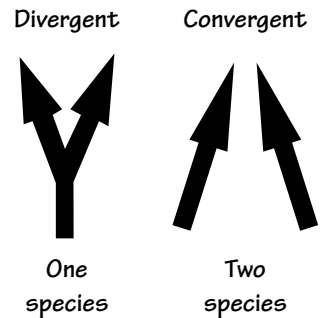
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## Divergent and convergent evolution

In divergent evolution one species splits into two groups that are initially very similar. (To *diverge* means to 'move apart'.) As time goes on, the two related species gradually evolve (due to environmental pressures) and develop more and more differences.

In convergent evolution, organisms that are not related to each other gradually develop similar characteristics and occupy similar niches in different ecosystems. (To *converge* means to 'come together'.)



- 10 What is meant by a *niche* in an ecosystem?
- 11 Whales are mammals descended from terrestrial animals, but they resemble fish. They have developed fins. Do whales share a close relationship with fish? Explain your answer.

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After his voyages, Darwin spent many years doing further study and refining his theories. He was reluctant to publish until Alfred Russell Wallace (who had been studying wildlife in the Malay Archipelago) sent him a draft of his work in 1858.

- 12 Discuss the social and political influences that might have made Darwin reluctant to publish his work on evolution.

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Because Russell's ideas so closely mirrored his own, Darwin was downhearted at first. He was eventually persuaded to deliver a joint paper with Russell in 1858.

- 13 Placental and marsupial moles are not related, but have similar habits and occupy similar habitats. What is the name of this phenomenon?

---

- 14 Write each of these terms beside its correct definition: *divergent*, *evolution*, *natural selection*, *marsupial*, *phenotype*.

- a** The type of evolution where different species arise from a common ancestor\_\_\_\_\_
- b** Physical characteristic of an organism\_\_\_\_\_
- c** The theory that organisms originated from other earlier types of organisms and have changed over generations\_\_\_\_\_
- d** The order of mammals that does not have a true placenta\_\_\_\_\_
- e** The reduction in the number of members of a population that are not as well suited to the environment as other individuals\_\_\_\_\_



Biogeography involves the study of the geographical distribution of animals and plants. This branch of science is used to investigate why organisms (plant or animal) are found where they are, and how that affects the land and the other species with which they interact.

Organisms have changed in many ways over long periods of time. Organisms change (evolve) according to Darwinian Theory, in response to changes in their environment. The physical location of an organism will affect its chances of survival and how it develops, if it survives.

- 1 In your own words, explain what is meant by 'the physical location of an organism'.

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Two people who have strongly influenced our thoughts on evolution are Charles Darwin (1809–1882) and Alfred Russell Wallace (1823–1913). In 1839, Darwin published his first accounts of his discoveries from his trip on *HMS Beagle*. He had looked closely at fossils and compared them with living animals in South America and the Galapagos islands. Darwin didn't put forward in detail his long-held beliefs on evolution until 1859, when he saw Wallace's almost identical propositions.

- 2 Suggest reasons why Darwin might have waited 20 years between forming his ideas and letting them become public knowledge.

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While in South America, Darwin found fossils of extinct armadillos that had differences from living animals. He also noted that rheas (flightless birds distantly related to emus and ostriches) varied throughout the South American countries. One of his major observations involved the fauna of the Galapagos Islands (off the western coast of Ecuador).

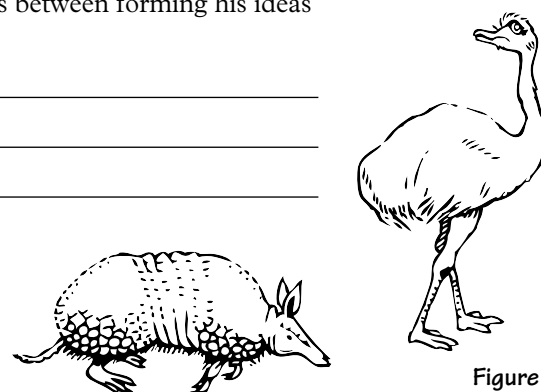


Figure 1

Figure 2

- 3 Define *extinct*. \_\_\_\_\_
- 4 Define *fossil*. \_\_\_\_\_
- 5 Define *fauna*. \_\_\_\_\_

Darwin noticed that birds and tortoises on the mainland of South America resembled those of neighbouring islands, whereas the equivalent animal populations of the nearby Galapagos Islands had significant differences, despite being the same species.

- 6 What is meant by *equivalent animal populations*? \_\_\_\_\_

---

Wallace spent several years in the mid 19<sup>th</sup> century exploring the interior of South America and then the Malay Archipelago (a group of islands). He noticed that there were similarities between living organisms and those that had died out. He stated that: 'Every species has come into existence coincident both in space and time with a pre-existing, closely allied species'.

- 7 Would the geography of an ecosystem affect the distribution of species? Explain your answer.

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- 8 *A science that deals with the geographical distribution of animals and plants* – what does this define?

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Biogeography looks at the locations of flora and fauna. In addition, biogeographers study the reasons why organisms are where they are.

- 9 Define *flora* and *fauna*. \_\_\_\_\_

---



The type and number of organisms of a particular species in a particular location will depend on the:

- previous climate
- present climate
- historical numbers of organisms of a species
- adaptability of the organisms.

**10** Define *climate*. \_\_\_\_\_  
\_\_\_\_\_

**11** What is meant by 'historical numbers of organisms of a species in a particular location'?  
\_\_\_\_\_  
\_\_\_\_\_

Joseph Dalton Hooker (1817–1911) was a contemporary of Darwin and Wallace. He made many botanical discoveries. He attempted to explain the geographical distribution of plants and why there were apparently anomalous variations.

**12** Define *contemporary*. \_\_\_\_\_

**13** Define *botanical*. \_\_\_\_\_

**14** Define *anomalous*. \_\_\_\_\_

Hooker was one of the scientists who supported the theories on *natural selection* advanced simultaneously by Darwin and Russell. He was one of the first people to study the discipline now known as biogeography.

Many regions have very distinctive flora and fauna. Australian eucalypts and marsupials are often quoted in theories of evolution. Because Australia has been isolated from the rest of the world for so long, organisms here have developed in a different way compared to other locations. Continental drift is used to explain why there are exceptions and why very similar organisms are found huge distances apart.

**15** Explain what is meant by 'continental drift', and how it might affect or direct evolution. \_\_\_\_\_  
\_\_\_\_\_

**16** Identify the following statements as true or false, then correct the false statements so that they are true.

- a** Charles Darwin was the only person in the 19<sup>th</sup> century who was investigating the theory of natural selection. \_\_\_\_\_
- b** Animals have adapted to changing conditions in their environments. \_\_\_\_\_
- c** All organisms developed from other organisms that already existed. \_\_\_\_\_
- d** Marsupials are native only in Australia. \_\_\_\_\_
- e** The study of fossils can prove evolutionary theories. \_\_\_\_\_

**17** Choose a plant or animal species that is known to exist in a specific area. Discuss reasons why that species might be found where it is (and not elsewhere).  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

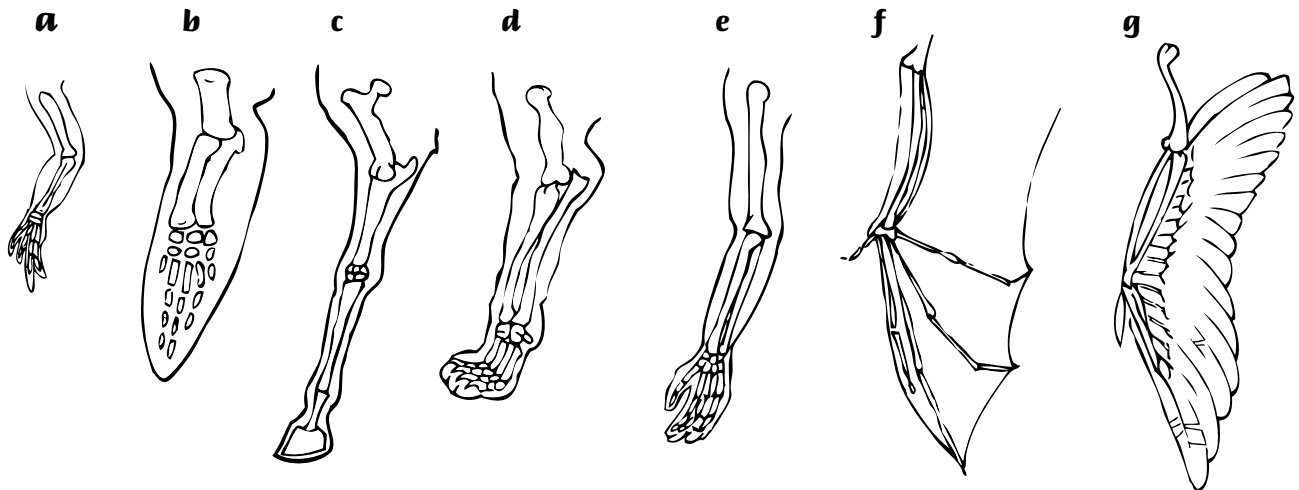
**18** Outline some of Sir Joseph Dalton Hooker's achievements.  
\_\_\_\_\_  
\_\_\_\_\_

**19 Research** – 'Wallace's Line' is often mentioned in relation to biogeography. Outline what this is.  
\_\_\_\_\_  
\_\_\_\_\_



The outward appearances of developing embryos and their body structures are often similar, even when the organisms are very different species. This is another example of nature being conservative. The forelimbs in many vertebrates (e.g. bat, human and whale) are *pentadactyl*; that is, have five 'fingers' (*penta* meaning 'five'). The forelimbs of these animals developed from the corresponding structures of primitive reptiles.

- 1** The diagrams below show the bones in the forelimbs of seven different animals. Label each diagram with the correct animal: *bat*, *bird*, *frog*, *horse*, *human*, *lion*, *whale*.



- 2** Write the name for the forelimb of a:
- a** human. \_\_\_\_\_
- b** whale. \_\_\_\_\_
- c** bat. \_\_\_\_\_
- 3** When different organisms have similar structure or physiology, because they have descended from a common ancestor – what does this define?

- 4** Which part of the human body could be compared to the wing and the flipper? \_\_\_\_\_

- 5** In what ways have the 'hands' of bats evolved differently from those of humans?

Comparative anatomy lends strong support to Darwin's Theory of Evolution. Darwin stated that new species evolve from pre-existing species (ancestors), by making gradual changes to fit their environments.

Most birds have wings, but some birds have adapted to an aquatic lifestyle.

- 6** Suggest why an animal that is commonly found in the skies might adapt to an aquatic lifestyle.

- 7** Name a species of bird that has done this. \_\_\_\_\_

- 8** What major adaptation occurred in the species you named?

- 9** All vertebrate embryos pass through a fish-like stage in which gills are present. What two things does this suggest about vertebrates?



- 10** Do an Internet image search for the 'comparative evolution of marine invertebrates'. You will see examples of great diversity and beauty. Sketch three examples in the box below.

- 11** Now do an Internet image search for the 'comparative evolution of vertebrate species'. Compare what you find with what you saw in the previous image search. What is the main difference between the two groups?

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## The Tumbler Pigeon

The tumbler pigeon is a breed of pigeon which is selectively bred for its 'ability' to perform spectacular tumbles and somersaults in mid air. These birds tumble when they stop flying in mid air, folding their wings close to them. They turn somersaults for a few seconds, dropping up to a metre, before resuming their normal flight. The tumbling is caused by the bird experiencing short epileptic seizures. When the seizure is over, normal flight resumes. Severely epileptic chicks seldom survive or do not survive long. Those that grow to adulthood, need their seizures monitored and treated with medication.

- 12** *Tumbler pigeons are an example of an exception to 'normal' evolution.* Is this a true statement? Explain your answer.

---

- 13** There are at least 10 types of tumblers bred across the world. Do you believe it is right to breed a bird (or any other animal) for its seizures, in order to create an 'amusing tumbler' for humans to own and show? Give reasons for your answer.

*Note:* There is no right or wrong answer – you are asked to explain your opinion.

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- 14** Find two examples of where human intervention has, through selective breeding, produced an animal species that would not survive if it were left up to natural selection. Explain why you believe natural selection would select against the examples you provide.

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- 15** Identify the following statements as true or false. Then correct the false statements so that they are true.

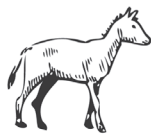
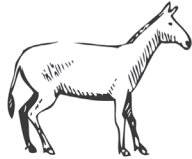
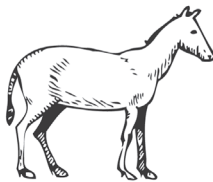
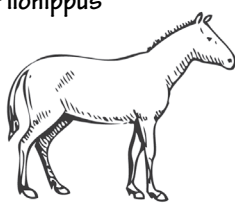
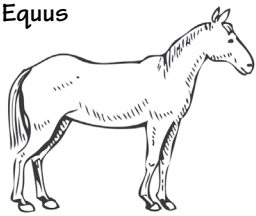
- a** Closely related organisms show similar patterns of development. \_\_\_\_\_
- b** The early stages of embryos of live-born animals (e.g. dogs) resemble early stage embryos of animals that develop in eggs (e.g. chickens). \_\_\_\_\_
- c** The early stages of embryonic development conclusively identify the type of organism growing. \_\_\_\_\_



Around 60 million years ago, there existed a four-toed animal about the size of a dog. Scientists have named this animal *Hyracotherium*. *Hyracotherium* is thought to be the ancestor of the modern horse (*Equus caballus*). Possible stages of the evolution of the horse are shown below.

Carefully study the changes in size and proportions, especially in the leg and molar teeth. Note that only some of the stages are included.

Fossil evidence can be interpreted in many ways and not all scientists agree on the sequence shown here.

| Time        | 60 million years ago                                                                               | 40 million years ago                                                                            | 25 million years ago                                                                             | 10 million years ago                                                                             | 3 million years ago                                                                          |
|-------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Name        | Hyracotherium<br> | Mesohippus<br> | Merychippus<br> | Pliohippus<br> | Equus<br> |
| Leg         |                  |               |                |              |          |
| Molar tooth |                 |              |               |             |         |

1 Referring to the diagram above, describe the changes that have occurred over time in:

- a** size. \_\_\_\_\_
- \_\_\_\_\_
- b** body proportions. \_\_\_\_\_
- \_\_\_\_\_
- c** legs. \_\_\_\_\_
- \_\_\_\_\_
- d** molar teeth. \_\_\_\_\_
- \_\_\_\_\_

2 Suggest possible reasons why only some stages in the horse's evolution are shown. \_\_\_\_\_

\_\_\_\_\_

The evolution of the horse occurred over a period of about 55 million years, and was not a linear event.

3 What is meant by a *linear* event? \_\_\_\_\_



Note that sources will vary regarding dates. For example, some sources list *Equus* as dating from about 5 million years ago to the present, and *Meryhippus* dates range from 18–11 million years ago. This sort of variation is to be expected when working with fossil evidence, so do not let it confuse you. The important thing to note is the general order in which scientists agree that the changes took place, and roughly when these changes are believed to have occurred.

- 4 Research** – The horse is believed to share a common ancestry with the rhinoceros and the tapir. This is because they share many similarities in their evolution. All three species have similar tooth and upper-lip structures. They, and the zebra, are classified as *perissodactyla*.

**a** What does *perissodactyla* mean? \_\_\_\_\_

**b** How does *perissodactyla* differ from *artiodactyl*? \_\_\_\_\_

- 5** Give one example of an animal classified as *perissodactyla*, and one example classified as *artiodactyl*.

- 6** The llama, like the animals named above, is an ungulate. Ungulates are defined as ‘hoofed animals’. However, not all ungulates have a common ancestry. Suggest how animals could have similar physical characteristics, yet not be related by a common ancestor.

- 7** Is the llama an *artiodactyl* or a *perissodactyl*? \_\_\_\_\_

- 8 Research Challenge – This is a tough one!**

Most ungulates are herbivores (grazers) and some are omnivores (eat plant or animal matter). Very few are carnivores. The extinct order *mesonychid* was a group of carnivorous ungulates. Two species of animal which exist today are believed to share a common ancestry with the *mesonychid*. Find out what these animals are, and then give the order to which these animals belong.

**Hints:** They are aquatic species that have triangular teeth, similar to those of the extinct carnivores (and no, they are not sharks!).

Horses and humans have had a close bond for a long time. Up until just over a century ago, humans relied on horses as part of their daily life.

- 9** List three ways in which people once relied on horses. \_\_\_\_\_

- 10** What sorts of characteristics did people breed horses for in those days, and why?

- 11** People still breed horses today, keeping accurate records on bloodlines, and selling many for ‘top dollar’.

**a** List two ways in which humans still share close bonds with the horse.

**b** Discuss why horses will always be a part of people’s lives.

- 12** Suggest a way that we could direct the evolution of horses.



Organisms are affected by their environment. The environment (surroundings) of an organism is made up of biotic and abiotic factors. Biotic factors are the living things that exist in the environment (*bios* meaning 'life'). Abiotic factors are non-living (*a-* meaning 'not', *a + bios* = 'not of life').

**1** Identify each of the following factors as biotic or abiotic.

- |                           |                                       |
|---------------------------|---------------------------------------|
| <b>a</b> temperature_____ | <b>h</b> minerals_____                |
| <b>b</b> rainfall_____    | <b>i</b> nesting sites in trees_____  |
| <b>c</b> food_____        | <b>j</b> nesting sites in cliffs_____ |
| <b>d</b> offspring_____   | <b>k</b> sunlight_____                |
| <b>e</b> predators_____   | <b>l</b> oxygen_____                  |
| <b>f</b> temperature_____ | <b>m</b> carbon dioxide_____          |
| <b>g</b> altitude_____    | <b>n</b> topography_____              |

Aquatic environments are those based in water, and terrestrial environments are on land (*aqua* meaning water, *terra* meaning earth). All animals move, although it is not so obvious with some species, such as the sea cucumber. Different creatures use different means to move through air, land and water.



Figure 1

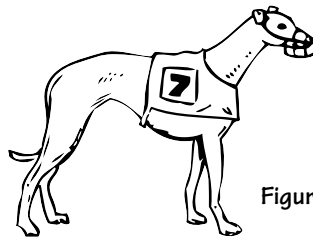


Figure 2

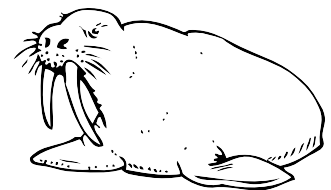


Figure 3

**2** Describe how the fish shown in figure 1 is suited to travelling quickly through water.

\_\_\_\_\_

**3** Describe how the dog shown in figure 2 is suited to travelling quickly on land.

\_\_\_\_\_

The walrus (figure 3) is a mammal that moves on land and in water. It is more mobile in water than on land because of the buoyancy of water. Water exerts an upward force, and the large mass of the walrus is supported when it is in the water.

**4** Name two other mammals that live on land but are more 'at home' in water.

\_\_\_\_\_

**5** If a species of animal seems to be clumsier on land than in water, why might it make its permanent home on land?

\_\_\_\_\_

\_\_\_\_\_

**6** Explain why jellyfish (coelenterates) do not keep their shape when taken from the sea.

\_\_\_\_\_

**7** *The organisms in an area that react with each other and their surroundings* – what does this define?

\_\_\_\_\_

**8** Identify the following statements as true or false.

**a** If an animal is *arboreal*, it lives in trees. \_\_\_\_\_

**b** An organism's *habitat* is the area in which it lives. \_\_\_\_\_



Temperatures in terrestrial environments vary much more than in aquatic environments.

**9** Write a definition for each of the following terms.

**a** terrestrial \_\_\_\_\_

**b** aquatic \_\_\_\_\_

Because fluctuations in temperature in bodies of water are less than the temperature fluctuations on land, life has frequently found it easier to exist in water. Life is thought to have originated in the oceans. Water is a much better conductor of heat than air is. Mammals and birds, when placed in water, lose body heat much more quickly than they do in air. Mammals and birds that spend time in cold water have some form of insulation to prevent heat loss.

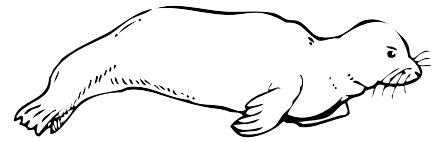


Figure 1

**10** Describe the insulation that mammals such as seals and sea lions have, to enable them to live in very cold water.

\_\_\_\_\_

**11** Hypothermia can be defined as a 'subnormal temperature of the body'. Explain why a person who is in a large body of water for a long period of time may suffer from hypothermia.

\_\_\_\_\_

**12** Outline the major differences between birds and mammals. \_\_\_\_\_

\_\_\_\_\_

Two main gases are involved in the metabolisms of living creatures – oxygen ( $O_2$ ) and carbon dioxide ( $CO_2$ ). Aerobic respiration is the process by which animals 'burn' food, using oxygen to produce heat energy and carbon dioxide (and other products).

Green plants use light energy, carbon dioxide and water ( $H_2O$ ) to produce oxygen and glucose ( $C_6H_{12}O_6$ ). The atmosphere contains about 20% oxygen and 0.03% carbon dioxide.

**13** Name one other substance that plants need for the process of photosynthesis. \_\_\_\_\_

**14** Do rivers and seas contain the gases oxygen and carbon dioxide? Explain your answer.

\_\_\_\_\_

**15** Light is used by plants for photosynthesis. What is one use that animals make of light?

\_\_\_\_\_

**16** There are many plants that have adapted to grow under water, such as seaweed. Suggest reasons why these plants tend to grow fairly well near the surface, but don't grow as well as the depth of water increases.

\_\_\_\_\_

**17** *Occurs only in the presence of oxygen* – what does this define? \_\_\_\_\_

**18** Write the formula for carbon dioxide. \_\_\_\_\_

**19** Write the formula for glucose. \_\_\_\_\_

**20** Write a word equation for the process of photosynthesis.

\_\_\_\_\_

**21** Explain in your own words what is meant by 'fluctuations in temperature in bodies of water are less than temperature fluctuations on land'.

\_\_\_\_\_

\_\_\_\_\_



An ecosystem consists of communities of organisms and the physical environment in which they live. (Remember, *populations* are groups of individuals belonging to the same species, and *communities* are the sum of the different populations living in a given area.)

All life on Earth is dependent on energy from the Sun. Green plants produce energy by photosynthesis. They are autotrophs; that is, they produce their own food from non-living (inorganic) components.

**1** Write another name for autotrophs. \_\_\_\_\_

Animals, fungi and most micro-organisms are unable to use directly the molecules that are present in non-living substances. They rely on plants to make complex molecules. If these organisms feed directly on green plants (that is, are *herbivores*), we call them *primary consumers*.

**2** What do we call an animal that eats a primary consumer? \_\_\_\_\_

A food chain is the sequence of organisms through which energy (and materials) flow in a community. Figure 1 shows a simple food chain. The arrow means 'is eaten by' or 'energy is transferred to'.

**3** In this food chain what is the:

**a** producer? \_\_\_\_\_

**b** tertiary consumer? \_\_\_\_\_

Ecologists describe the successive levels of nourishment in a food chain as *trophic levels*. Producers make up the first trophic level; primary consumers are the second trophic level, and so on. As you progress up through trophic levels, generally the organisms become larger and less numerous.

Food chains do not exist in isolation. They form complex linkages called *food webs*. Figure 2 shows a simple food web.

**4** Does this food web in figure 2 show all of the producers and consumers involved with these organisms? Explain your answer.

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**5** What is signified by the arrows in the food web? \_\_\_\_\_

**6** How many trophic levels are there in this food web? \_\_\_\_\_

**7** Write a definition for *population*. \_\_\_\_\_

**8** Write a definition for *community*. \_\_\_\_\_

**9** As organisms are eaten by higher order consumers, is all of the energy passed on to the next trophic level? Explain your answer.

---



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Every food chain or web begins with autotrophic (producer) organisms and ends with decomposers. *Autotroph* means 'self nourish'. Plants are autotrophs. They produce their own food without consuming another living organism.

Figure 3 shows the relationship between various trophic levels in an ecosystem.

Green plants → mice → snakes → bird of prey

Figure 1

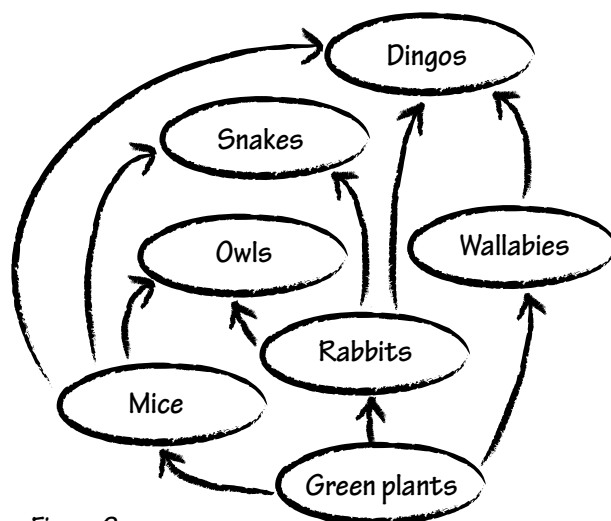


Figure 2

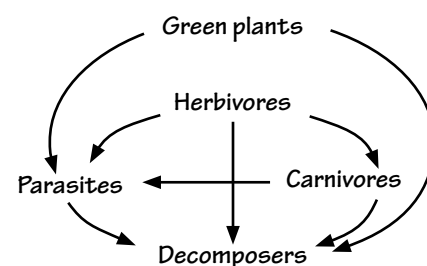


Figure 3





An *environment* (surroundings) is a combination of the living things, physical features and geological features in a particular area. We tend to divide environmental factors into *biotic* and *abiotic* features. Living creatures are described as biotic (*biotikos* meaning 'to do with life'). Things like sunlight and the amount of water present are abiotic factors – i.e. without life (*a-* meaning 'not').

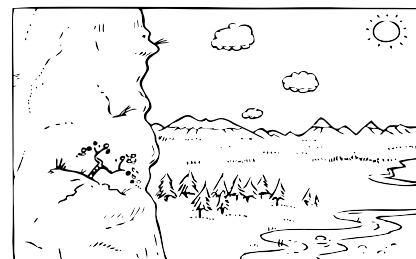


Figure 1

- 1 Name other abiotic factors of an environment.

\_\_\_\_\_

The area in which an organism lives is its *habitat*. Habitats can be aquatic, terrestrial or arboreal.

- 2 Write definitions for the following terms.

**a** aquatic \_\_\_\_\_

**b** terrestrial \_\_\_\_\_

**c** arboreal \_\_\_\_\_

- 3 For each of these habitats, list three examples of organisms that can be found living there.

**a** aquatic \_\_\_\_\_

**b** terrestrial \_\_\_\_\_

**c** arboreal \_\_\_\_\_

- 4 A student used these words as synonyms for habitat: *home*, *locale*, *surroundings*, *territory*. How suitable are these words? Explain your answer.

\_\_\_\_\_  
\_\_\_\_\_

Living creatures can be divided into two major groups, *flora* and *fauna*. Flora (*floris* meaning 'flower') refers to plants, and fauna refers to animals.

- 5 Name a plant that grows under water. \_\_\_\_\_

There are different names used to describe groups of organisms. A *species* is a group of related organisms that are capable of breeding to give fertile offspring.

- 6 What is meant by *fertile offspring*? \_\_\_\_\_

A *population* is the number of a particular species in a habitat at a given time. For example, if we count all the people living in Australia, that number is Australia's population. However, if we only count the people living in NSW, that number is the population of NSW.

A biological *community* is all of the organisms (of many different species) in a location, and how they interact with each other. Each successful organism usually occupies its own *niche*, which refers to the place the organism inhabits, and the role it plays in the *ecosystem* to which it belongs.

An *ecosystem* is all of the organisms in an environment, and the relationships between the animals and plants themselves and with their physical environment.

The *distribution*, *diversity* and numbers of plants and animals in ecosystems are determined by the biotic and abiotic factors.

- 7 Explain the difference between a *population* and a *community*. \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_

- 8 Write definitions for the following terms.

**a** distribution \_\_\_\_\_

**b** diversity \_\_\_\_\_



- 9** List as many differences as you can between the abiotic components of terrestrial environments and those of aquatic environments.

| Terrestrial environments | Aquatic environments |
|--------------------------|----------------------|
|                          |                      |

- 10** Choose two animals and describe the different ways in which they might interact with their environment.

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- 11** Many people think that all fish are the same, or all birds are the same, in terms of how they interact with their environments. Do you agree or disagree? Explain your answer, using either fish or birds as an example.

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- 12** Describe ways in which your local environment might differ from each of the following:

**a** A tropical island community (e.g. New Guinea)

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**b** A northern hemisphere community (e.g. Alaska, northern England)

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- 13** Describe the native plants found in your local community. Could they be found anywhere across Australia? Explain your answer.

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- 14** Describe the exotic (introduced) plants found in your local community. Could they be found anywhere across Australia? Explain your answer.

---

- 15** Draw a concept map or mind map to summarise what you found out in this worksheet. Include the terms: *environment, biotic, abiotic, habitat, arboreal, terrestrial, aquatic, population, community, distribution, diversity*.

Local environment



Exobiology (also called astrobiology) is the study of the origins, evolution and distribution of life within our universe, not just on the planet Earth (*exo* meaning ‘outside’). NASA’s exobiology program is an example of one of several organisations actively investigating this mystery. The interest taken by institutions like NASA has given the field a new respectability. It is believed the question of the origins of life can best be answered by dividing it into the study of the four main parts listed below:

- Prebiotic evolution (*pre* meaning ‘before’, *biotic* meaning ‘living/life’)
- The evolution of biogenic compounds (How did they originate?)
- Early evolution of living organisms (e.g. How did life arise and evolve?)
- Evolution of more advanced organisms (e.g. Does life exist on other planets? If so, is it more/less advanced than life on Earth?)

Note that other groupings than the above have been made, but we will only consider the above.

The main elements of life are carbon, hydrogen, oxygen, nitrogen, phosphorous and sulphur (commonly abbreviated as CHONPS). We usually think of these elements as making up Earth’s life forms, but they are also part of the make-up of the planets, stars and other space matter. Because of this, there is every reason to believe that other life forms exist in our universe, since it is so large.

Scientists have traced the origins of the elements as far back as coming from stars. From there, they went on to form planets and other space matter. What we don’t know is where the elements came from in the first place, in order for the stars to be formed. This is one of the many mysteries scientists are trying to solve. With so many different compounds in existence, there are many pathways to investigate.

- 1** How would the work of an exobiologist/astrobiologist differ from the work of an astrophysicist?

---

Scientists use life on Earth as a guide to how they will investigate the origins of life in the universe.

- 2** Suggest why they do this. \_\_\_\_\_

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- 3** List some advantages and disadvantages of doing this. \_\_\_\_\_

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Chemistry is the study of the composition and behaviour of substances. Two major branches of chemistry are organic chemistry (carbon-based compounds) and inorganic chemistry (non-carbon-based compounds). Since life as we know it is based on carbon, exobiologists begin their investigations with the assumption that other life forms will also be carbon-based (therefore organic in origin). However, before the carbon-based life forms appeared, the environments in which they evolved had to be created. This presents a whole new set of questions which may not necessarily involve organic chemistry exclusively.

- 4** Suggest factors that would affect how the environments in which life forms evolved would themselves form.

---

- 5** A biogenic substance is a substance that is produced as a result of a plant or animal’s natural life processes. Silk and coal are examples of biogenic substances. Name two others.

---

Exobiology is a multidisciplinary branch of science. Biology, chemistry, geology, physics and astronomy all play significant roles.

- 6** Suggest reasons why all these branches of science might be needed in exobiology.

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- 7** Although exobiology is multidisciplinary, there may be times when investigations are oriented more to a specific branch of science, such as biology or geology. Discuss why this may happen.

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- 8** Some people argue that exobiology is not a pure science in the same way that biology, chemistry and physics are. Outline the points for and against this argument. Discuss your answers.

| For                                      | Against                                  |
|------------------------------------------|------------------------------------------|
| <br><br><br><br><br><br><br><br><br><br> | <br><br><br><br><br><br><br><br><br><br> |

- 9** Methane gas ( $\text{CH}_4$ ) is produced by many living things. It is broken down by ultraviolet radiation. Methane gas was found in the Martian atmosphere. Scientists calculated that on Mars, methane is broken down very quickly. This caused some excitement, as it left people wondering how methane got there, suggesting some form of life might be responsible. What do you think may be the source?

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- 10** Another assumption made by exobiologists is that water is involved in the evolution and formation of life forms. Why is this a reasonable assumption to make?

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Most forms of life on Earth need heat and light to survive.

- 11** How would these things be found elsewhere in the universe?

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- 12** What sorts of assumptions could be made about heat and light in the universe?

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- 13** Write a definition for *cosmic evolution*.

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- 14** Outline how each of these branches of science might help us to understand the evolution of life forms.

**a** Geology\_\_\_\_\_

**b** Biology\_\_\_\_\_

**c** Physics\_\_\_\_\_

**d** Chemistry\_\_\_\_\_

**e** Astronomy\_\_\_\_\_

- 15** Discuss this statement: *Because we haven't found conclusive evidence of life off Earth, exobiology is a waste of time, and all our resources should be spent on studying human life.*

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Ants (and all other organisms) are classified using the following system.

- **Kingdom:** Animalia
- **Phylum:** Arthropoda
- **Class:** Insecta
- **Order:** Hymenoptera
- **Family:** Formicidae

1 What are the next two major levels after *Family*?

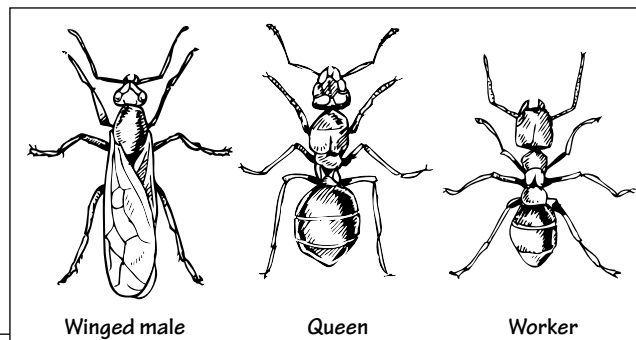


Figure 1

2 Read the following passage and write definitions for the terms in *italics*.

There are nearly 4000 different *species* of ants. Ants are very social organisms and live in large populations. Some types of ants create large mounds (ant hills), usually made of soil. There are three major *castes* of ant: males, workers and queens. Males hatch from unfertilised eggs (are *haploid*) and females hatch from fertilised eggs (are *diploid*). Worker ants are (usually) *sterile* females. Workers are the most numerous ants in a colony and carry out many roles such as farmers that tend aphids, *scavengers* and *predators*.

- a** species \_\_\_\_\_
- b** haploid \_\_\_\_\_
- c** diploid \_\_\_\_\_
- d** sterile \_\_\_\_\_
- e** scavenger \_\_\_\_\_
- f** predator \_\_\_\_\_
- g** caste \_\_\_\_\_

A researcher studied how the walking speed of ants varied with soil temperature. The results are in the table below.

| Temperature (°C)              | 5   | 10   | 15  | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|-------------------------------|-----|------|-----|------|------|------|------|------|------|------|
| Speed (cm sec <sup>-1</sup> ) | 1.5 | 1.75 | 2.0 | 2.25 | 2.50 | 2.70 | 2.80 | 2.90 | 2.40 | 1.80 |

3 On graph paper, plot a line graph using the figures in the table above. Plot temperature on the x-axis and speed on the y-axis. Label the axes and include the units used.

4 Forecast the likely walking speeds of ants at:

- a** 0°C \_\_\_\_\_
- b** 55°C \_\_\_\_\_

5 Suggest other factors that might affect the speed at which ants move. \_\_\_\_\_

6 What else can you deduce from reading your graph? \_\_\_\_\_

When a worker ant finds food, she secretes a pheromone onto the soil as she returns to the nest. Other workers can follow the trail. This substance is volatile; that is, it evaporates quickly. The workers stop laying the pheromone when the food runs out. This means that workers do not waste time travelling to an exhausted food supply.

7 Each species of ant has its own, unique pheromone. Why is this? \_\_\_\_\_

8 A chemical substance excreted by an organism into the environment, that other organisms of the same species can sense and that causes their behaviour to change – what does this define? \_\_\_\_\_

9 Ant species are polymorphic. Define *polymorphic* and include an example relating to ants.



A bacterium is a single-celled (unicellular) organism. The plural of *bacterium* is *bacteria*. There are many ways to classify bacteria, such as by their shape or by the environment in which they are found.

Usually when people hear the word 'bacteria' they think of germs that can cause illness. However, some bacteria are helpful. Bacteria can be used to make cheese and yoghurt and medicines such as antibiotics. Bacteria are microscopic organisms. They are unicellular and reproduce by fission; that is each individual cell splits into two (identical) 'daughter' cells. This type of reproduction is 'asexual', meaning it doesn't involve males and females, it doesn't require two parents. Because bacteria can reproduce so quickly, they can generate large numbers in a very short period of time.

Figure 1 shows how the numbers of a particular type of bacteria change over a period of time.

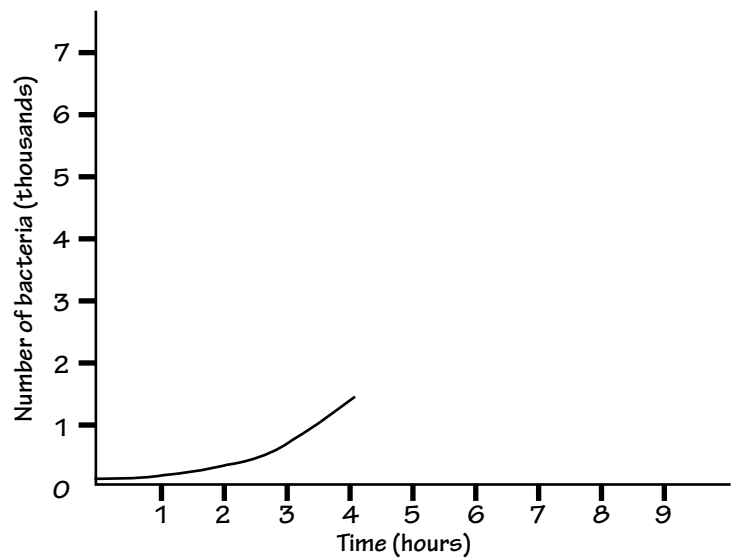


Figure 1

- 1 Extrapolate (continue the plot of the line graph) to show the likely numbers of bacteria in the incomplete part of the graph.
- 2 Showing your working on the graph, estimate the numbers of bacteria likely to be present at the following times.
  - a 0 hours \_\_\_\_\_
  - b 1 hour \_\_\_\_\_
  - c 4 hours \_\_\_\_\_
  - d 7 hours \_\_\_\_\_
  - e 9 hours \_\_\_\_\_
- 3 Do you think that the numbers of bacteria will continue to increase indefinitely as indicated by the graph? Explain your answer.

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- 4 Write definitions for the following terms.
  - a unicellular \_\_\_\_\_
  - b plural \_\_\_\_\_
  - c classify \_\_\_\_\_
  - d microscopic \_\_\_\_\_
  - e fission \_\_\_\_\_
  - f asexual \_\_\_\_\_
- 5 List one negative effect of bacteria. \_\_\_\_\_
- 6 List one positive effect of bacteria. \_\_\_\_\_
- 7 Antibiotics are used to kill bacteria. True or false? \_\_\_\_\_

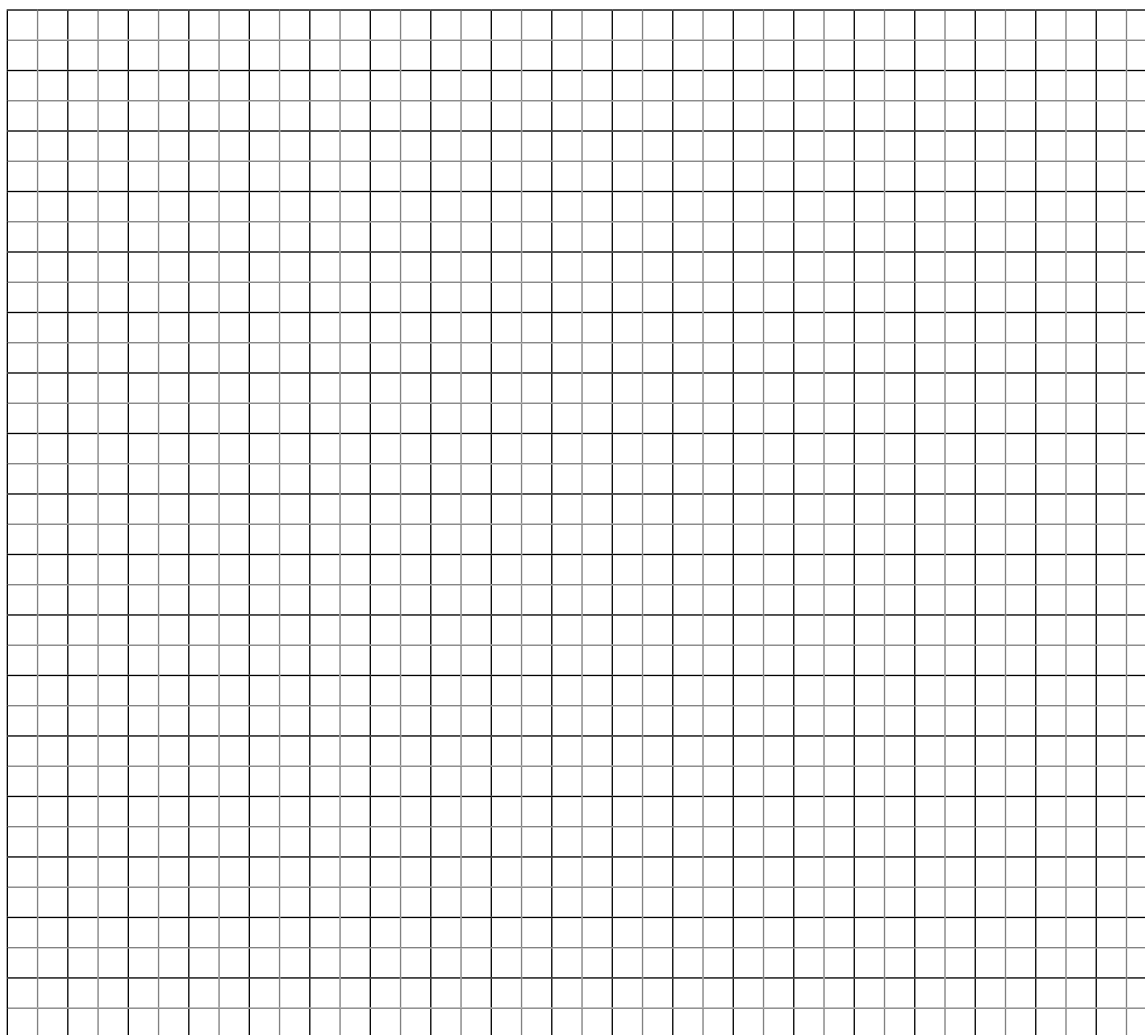


The common house mouse has the scientific name *Mus domesticus*. It is an introduced species. Mice can cause huge amounts of damage to crops, equipment and buildings. The cost is estimated to be hundreds of millions of dollars per year in Australia, increasing in times of plague. Mouse plagues occur because mice can breed very rapidly if food is plentiful and weather conditions are fine. Female mice give birth to a litter (on average, 5 young) less than 3 weeks after mating. They can become pregnant almost immediately after giving birth.

Sick mice can also spread disease. The table below shows the estimated density of mice per hectare (ha) over a period of time at a pastoral property in Victoria.

| Time (months)              | 0 | 1 | 2  | 3   | 4   | 5    | 6    | 7    | 8   | 9  | 10 | 11 | 12 |
|----------------------------|---|---|----|-----|-----|------|------|------|-----|----|----|----|----|
| Mouse density (No. per ha) | 2 | 5 | 80 | 350 | 825 | 1350 | 1400 | 1150 | 250 | 20 | 10 | 5  | 2  |

- 1 Plot a graph of mouse density ( $y$ -axis) against time ( $x$ -axis). Label the axes and include the units used.



- 2 Describe what can be observed about the population and then explain why the mouse numbers varied so dramatically.

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A human baby stays in the uterus (womb) for 38 weeks after conception. This is called the *gestation period*. Not all mammals have the same gestation period. In cats it is 8 weeks, dogs 10 weeks, elephants 83 weeks, mice 3 weeks and sheep 21 weeks.

- 1 Use figure 1 to show (as a bar graph) the gestation periods of the species mentioned.
- 2 Comment on any links between the length of the gestation period and the size of the adult animal.

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As a child grows older, the size of his/her body increases. However, the different parts of the body don't all grow at the same pace and at the same time. Figure 2 shows how the relative sizes of some body parts change over time, compared to birth size.

- 3 Which part changes the most over time?  

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- 4 Which part changes the least?  

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- 5 How long does it take the brain to reach its maximum size?  

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- 6 Study the line that shows the rate of growth of the heart. Is this rate likely to continue as shown? Explain your answer.  

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- 7 Consider a child that weighs 3 kg at birth (0 years). Using the graph in figure 2, predict his/her body weight at the following ages.

a 5 years \_\_\_\_\_

b 12 years \_\_\_\_\_

c 18 years \_\_\_\_\_

- 8 A child might not actually achieve the body weight predicted for their particular age. List three possible reasons for this.

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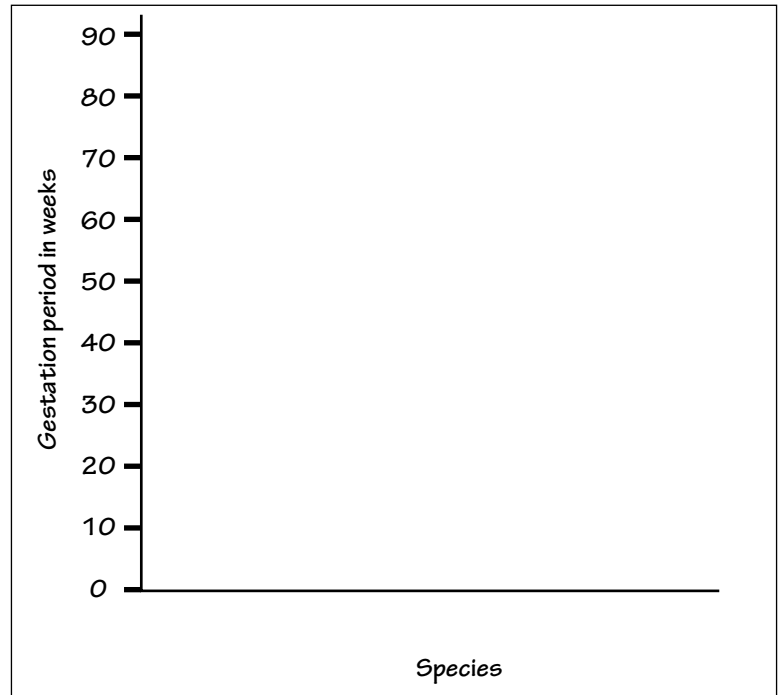


Figure 1

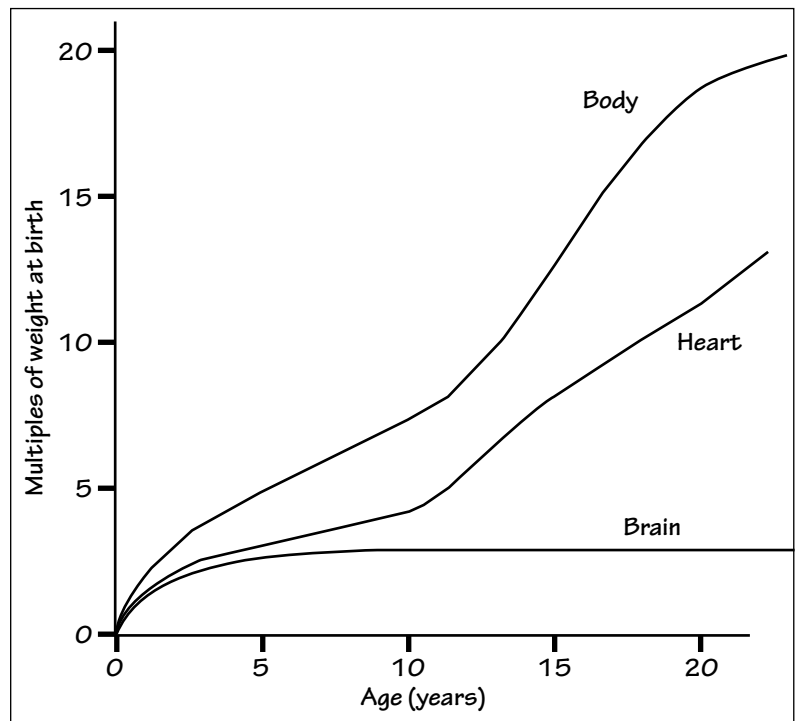


Figure 2



A class of ten-year-old students discussed how they would develop over the next few years. They carried out a survey to find out if there was a relationship between the heights of fathers and the heights of their sons.

Table 1 shows their results.

- 1 What is the minimum height of the fathers? \_\_\_\_\_
- 2 What is the minimum height of the sons? \_\_\_\_\_
- 3 What is the maximum height of the fathers? \_\_\_\_\_
- 4 What is the maximum height of the sons? \_\_\_\_\_
- 5 What is the average height of the fathers? \_\_\_\_\_
- 6 What is the average height of the sons? \_\_\_\_\_
- 7 In figure 1, draw a scatter gram (individual points) of the results from Table 1. Plot son's height on the  $x$ -axis and father's height on the  $y$ -axis.
- 8 Describe the pattern that is shown on your scatter gram.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- 9 List two other factors that might influence the final height of a son, apart from the height of the father.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- 10 How could you adapt this investigation to get more reliable data?  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- 11 How would you expect a similar study of mothers and daughters to turn out? Explain your answer.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- 12 Why do some populations increase or decrease in size? For example, *The Age* reported that Australians are growing taller. Another report stated that teenagers are, on average, 2 cm taller than teenagers of the same age in the previous generation.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Table 1

| Height of father (cm) | Height of son (cm) |
|-----------------------|--------------------|
| 177                   | 114                |
| 171                   | 128                |
| 163                   | 114                |
| 162                   | 116                |
| 172                   | 140                |
| 174                   | 132                |
| 179                   | 133                |
| 181                   | 143                |
| 180                   | 124                |
| 176                   | 131                |
| 185                   | 149                |
| 175                   | 124                |
| 177                   | 138                |
| 184                   | 147                |
| 178                   | 141                |
| 180                   | 128                |

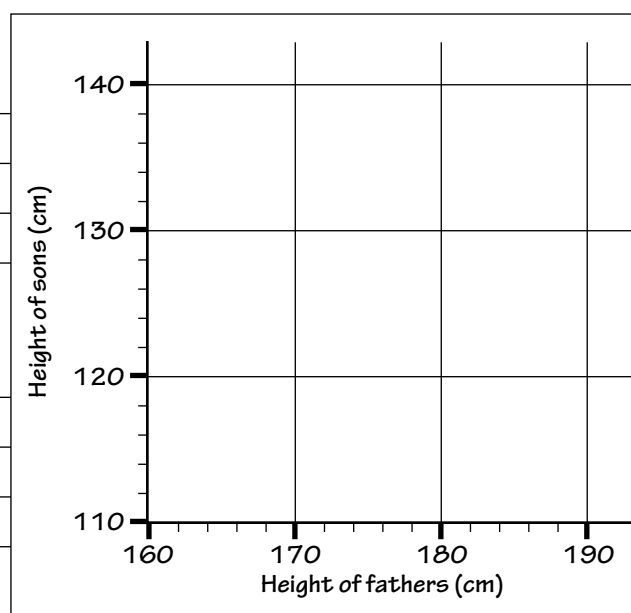
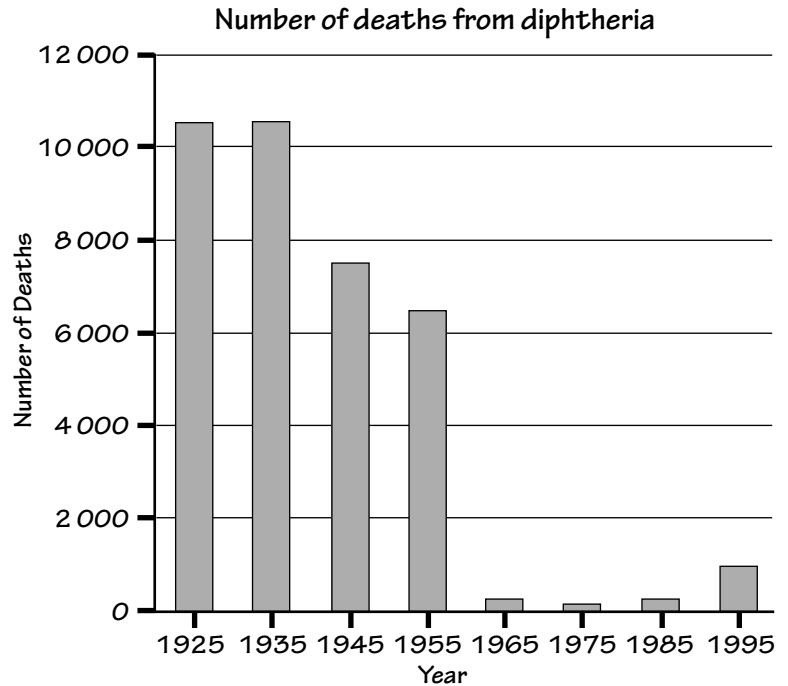


Figure 1



Diphtheria is a highly contagious disease that causes inflammation of the heart and nervous system. Prior to the advent of a specific antibiotic to treat this disease, the mortality rate for people contracting it was about 35%. Diphtheria was once widespread but with the advent of childhood vaccinations, the number of affected individuals has dropped dramatically. Antibiotics are only useful against bacteria, they do not affect viruses.

The bar chart shows the number of deaths due to this disease over a period of time in one particular country.



- 1 When was the antibiotic treatment first introduced? Give a reason for your answer.  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
- 2 When did the use of the antibiotic first control the disease? Give a reason for your answer.  
\_\_\_\_\_  
\_\_\_\_\_
- 3 Was this disease caused by a virus or a type of bacteria? Give a reason for your answer.  
\_\_\_\_\_  
\_\_\_\_\_
- 4 Make up another question that could be asked about the bar chart and information, and then answer it.  
\_\_\_\_\_  
\_\_\_\_\_
- 5 In recent years, scientists have been experimenting with new treatments for diphtheria. Why might they be doing this?  
\_\_\_\_\_  
\_\_\_\_\_
- 6 **Research** – Find out what the symptoms of diphtheria are.  
\_\_\_\_\_  
\_\_\_\_\_
- 7 What is meant by a '35% mortality rate'?  
\_\_\_\_\_
- 8 With a 35% mortality rate, roughly how many people would die out of a group of 250? \_\_\_\_\_
- 9 Explain what is meant by *contagious*. \_\_\_\_\_  
\_\_\_\_\_
- 10 Name one antibiotic in current use. \_\_\_\_\_
- 11 Find out what is meant by the term *epidemiology*. \_\_\_\_\_  
\_\_\_\_\_



# Selected answers



## 1 – Characteristics of living things

- a** Biologists **b** Characteristics  
**c** Excreting **d** Reproducing  
**e** Multicellular **f** Classify  
**g** Microbes
- Magnifies things
- How they move
- Live in water, have fins, eat fish
- Breathe air, give birth to live young, constant body temperature
- a** False **b** True **c** False  
**d** False **e** False **f** True  
**g** True **h** False **i** True

## 2 – Cell theory – A

- Too small to be seen with the naked eye
- Hollow space
- Both similar, spaces surrounded by walls, regular structure.
- Decayed over time
- Went against accepted theories, microscopes not common
- a** Organelle **b** Botany  
**c** Physiology **d** Nucleus  
**e** Cell **f** Pathology  
**g** Biogenesis
- a** First to observe and name cells  
**b** Observed nucleus in plants  
**c** Plants made of cells  
**d** Animals made of cells  
**e** Cells arise from existing cells  
**f** Invention of microscope

## 3 – Cell Theory – B

- Corroborate
- Proposition based on observation
- a** The process of making an image appear larger  
**b** Separates close objects from each other  
**c** Enables parts of cell to be distinguished
- Electron
- Gives better contrast and living cells and their processes can be observed
- The light microscope uses white (visible) light and glass lenses. The electron microscope uses a beam of electrons focused on a fluorescent screen
- A place where no matter is present
- So that other atoms don't interfere with the beam of electrons
- To increase the colour contrast between parts of cells
- Cheaper, simpler to use, still gives acceptable images, specimen

doesn't have to be killed

## 4 – Cell organelles

- Plant cell
- Animal cell
- a** Chloroplast **b** Vacuole  
**c** Cell wall **d** Nucleus  
**e** Cytoplasm **f** Cell membrane
- a** Golgi bodies  
**b** Endoplasmic reticulum  
**c** Ribosomes **d** Lysosomes  
**e** Mitochondria **f** Nucleolus
- Teacher to check, resemblance between organs and organelles
- Yes, e.g. bacteria

## 5 – Cell chemicals

- a** H<sub>2</sub>O **b** CO<sub>2</sub> **c** O<sub>2</sub>
- Carbon, hydrogen and oxygen
- Opposite charges attract them
- Input; more energy is needed to break bonds than is given out making bonds.
- Carbon
- Sodium, potassium and calcium
- Oxygen present
- Mono* made up of one unit, *di* – two units and *poly* – many units
- Monosaccharide because it has a simpler structure
- Contains carbon, hydrogen and oxygen. Hydrogen and oxygen are in the same ratio as in water
- Teacher to check, e.g. pasta, bread, cereal
- Releases energy more quickly than other foods
- For growth and repair of the body
- Teacher to check. Substance not based on the element carbon. Inorganic substances are those not originating from living or once living organisms, e.g. rocks
- C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>
- Top: store food, carbohydrate, respiration  
Bottom: carbon, hydrogen, oxygen, energy, carbon dioxide, water

## 6 – Endotherms and ectotherms

- An ectotherm, its body temperature fluctuates with air temperature
- 36°C
- Build up
- Flourish, do well
- e.g. lizard
- Because they need that much less

food, etc.

- Endotherms can be active for longer periods than ectotherms
- a** Heat lost, evaporating liquid  
**b** Reduce heat loss to surface  
**c** Avoid extremes of temperature

## 7 – Adaptation

- Should outline that the fittest survive to reproduce and pass on these characteristics to offspring. Natural selection relies on there being a variation in the population
- Water, mates
- e.g. birds, search for food, water, mates, better weather
- Physiology
- Selection has taken place. Cows grazed on taller, upright plants
- a** To do with the ground  
**b** To do with trees  
**c** Grasping
- Smell, shape
- Camouflage
- e.g. butterflies, moths
- Won't get eaten
- Attitude, teeth, claws
- Note that some characteristics selected by breeders do not have a survival function, but only appeal to the breeder
- Desert
- a** Stops it being eaten **b** Gets water deep below surface  
**c** Less water lost by transpiration
- a** Structural **b** Structural  
**c** Structural
- Teacher to check; could include growth of tubers, waxy leaves, bitter-tasting leaves
- Conserve water, avoid predators, find prey
- Physiological
- Shaped to minimise drag, e.g. like a teardrop
- Active at dawn and dusk
- Teacher to check; could include flight and defence mechanisms

## 8 – Competing species

- They will die out or migrate
- Not many other predators to keep its levels down. Very successful
- Eat crops, grazing land for cattle and sheep
- e.g. Many natural predators, weather etc., all keeps numbers down
- a** Produce large numbers  
**b** Able to breed **c** Birth group  
**d** Time from conception to birth



# Selected answers (cont'd)



- 6 Hooves are hard, sharp and animals are heavy. Cut into and break up top soil and vegetation
- 7 Herbivore numbers would initially increase to maximum due to drop in number of carnivores. Herbivore numbers would drop once all vegetation was exhausted
- 8 Water, space, sunlight
- 9 Ecosystem

## 9 – Natural selection

- 1 Increase
- 9 Gradually end up with only yellow beetles
- 10 Need some guidelines from which to draw conclusions
- 11 By chance
- 12 Beetles only die from predation, fertility rate stays the same, ratio of predation stays the same
- 13 e.g. make predictions about how things might change

## 10 – Evolution – A

- 1 Grow, eat, reproduce, excrete waste, respond to stimuli
- 2 **a** Carbon **b** Hydrogen **c** Oxygen **d** Nitrogen
- 3 Plants with chlorophyll use energy from sunlight to form oxygen and carbohydrates from water and carbon dioxide
- 4 Bacterium, larva
- 5 True
- 6 Connected with the sea
- 7 No internal skeleton
- 8 External skeleton or shell/casing
- 9 Teacher to check
- 10 Frog, toad, newt, salamander
- 11 Teacher to check
- 12 Teacher to check
- 13 Teacher to check
- 14 **a** False **b** True **c** True
- 15 Not necessarily. Depends on how well a species copes. Time scale also important
- 16 Charles Darwin

## 11 – Evolution – B

- 1 Traits, e.g. physical or behavioural
- 2 Ability to remain undetected, strength, teeth
- 3 Evolution
- 4 Roots
- 5 Leaves, and some photosynthesis takes place in the stems of green plants
- 6 **a** Occurs naturally in that area, not introduced **b** Energy from the Sun **c** Brought from

another place by humans  
**d** Cleaning agent **e** Take the place of **f** Flourish

- 7 **a** Endure, put up with **b** Higher than normal **c** Manage, deal with
- 8 Water
- 9–10 Teacher to check, e.g. an unusual plant might be a new food source if the animal can learn to digest it. An animal's constant grazing might lead to a plant developing thorns

## 12 – Evolution – C

- 1 Carbon dioxide, oxygen
- 2 Somewhere to live/shelter
- 3 b/c/d, g/i/e j k h. More than one sequence is possible
- 4 The vertebrae are very large. Those with the larger ones grew longer necks
- 5 Lots of plants, many high up
- 6 Large leaves, tall
- 7 Resource
- 8 Rain and sunlight

## 13 – Evolution – D

- 1 Imprints, solid rock type, bones
- 2 To see how the animal has changed over time
- 3 Rocks formed by the hardening of sediments, e.g. sandstone, shale and limestone
- 4a Ensure the layers are in correct geographical order
- b** Can be disturbed by earthquakes, mining, etc.
- c** History of events such as earthquakes, tremors; councils may know if building/excavation work has been done; stories and memories of local people.
- 5 Not all areas will have a complete or undisturbed record
- 6 As life has evolved, it has often become more complex
- 7 Not all life leaves fossils, some may have been destroyed
- 8 The fossils arranged in chronological order
- 9 Teacher to check
- 10 Teacher to check
- 11 Amphibians
- 12 Shows the change in an organism
- 13 Lungs, round fins, paired fins
- 14 e.g. In search of food, mates, curiosity, response to changes to existing environment, expanding territory due to overcrowding

## 14 – Darwin

- 1 Individual is the single organism,

population is all of the individuals grouped together

- 2 Population, it is populations that ensure the survival of a species, not a single individual
- 3a Teacher to check, e.g. same number of arms legs, eyes
- b** Teacher to check, e.g. eye colour hair, size
- 4 e.g. food, shelter, territory, mates
- 5 Type of food they could eat, fight with other birds
- 6 e.g. the largest of the large beaked get mates, and the most delicate of the small beaked get mates – this selection directs further beak selection
- 7 Not located near other places
- 8 Animals can change their characteristics over time
- 9 Teacher to check
- 10 Status or role of an organism in its environment
- 11 No, similarities are external
- 12 Went against accepted theories, religion
- 13 Convergent evolution
- 14 **a** Divergent **b** Phenotype **c** Evolution **d** Marsupial **e** Natural selection

## 15 – Biogeography

- 1 Where it lives
- 2 Wanted to be sure, lacked confidence, they went against accepted beliefs
- 3 No examples of that organism still living
- 4 Trace or remains of a previously living creature
- 5 Animals
- 6 The corresponding groups of similar animals of the same species
- 7 Yes, teacher to check that students understand that geographical features affect shelter, plant life, etc.
- 8 Biogeography
- 9 Plants, animals
- 10 Long-term pattern of weather
- 11 How many animals have lived there over time
- 12 From the same period of time
- 13 To do with plants
- 14 Inconsistent
- 15 Slow movement of tectonic plates resulting in land masses moving; land mass may move to a different climate, which would affect flora and fauna
- 16 **a** False **b** True **c** True



# Selected answers (cont'd)



**d** False **e** False

- 17** Teacher to check
- 18** Answers could include: botanist, encouraged Darwin and Wallace, travelled widely, discovered many new species, described and collected world flora
- 19** Answers could include: proposed by Alfred Russell Wallace, line between Australasia and oriental regions, marks limit of distribution of several species.

## 16 – Comparative anatomy

- 1** **a** frog **b** whale **c** horse  
**d** lion **e** human **f** bat  
**g** bird
- 2** **a** Arm **b** Flipper **c** Wing
- 3** Comparative anatomy
- 4** Hand
- 5** For example, five fingers, grew longer, skin membrane goes the length of the fingers to the tips.
- 6** Lives in cold regions where fish is the major food source
- 7** Penguin
- 8** Wings evolved into flippers
- 9** Likely a single common ancestor that originally lived in water
- 10** Teacher to check
- 11** Invertebrates have very different embryos whereas there are more clearly recognisable similarities between vertebrates
- 12** No, because they only survive as a result of selective breeding by pigeon breeders and reliance on medication
- 13** Teacher to check
- 14** Teacher to check
- 15** **a** True **b** True **c** False

## 17 – Evolution of the horse

- 1** **a** Got bigger **b** Remained similar **c** Longer, digits have merged **d** Bigger, longer
- 2** These are the major ones
- 3** One that occurs in a regular way, one after another
- 4** **a** Odd number of toes  
**b** Even number of toes
- 5** To be checked, for even toed answers include, zebra, horse, tapir, for artiodactyls, goat, cow, hippo
- 6** To be checked, e.g. the trait is of use to them
- 7** Artiodactyl
- 8** Killer whales and dolphins
- 9** To be checked, e.g. transport, pets, ploughing fields
- 10** e.g. strength and stamina
- 11** e.g. as pets, in the racing industry,

where much money is to be made, rounding up stock

- b** e.g. People have empathy with horses, horses will always be useful to people
- 12** e.g. for speed or stamina on the racetrack, for a physical appearance or to keep a breed alive, e.g. brumbies, mustangs, Indian painted, etc.

## 18 – Components of ecosystems

- 1** **a** Abiotic **b** Abiotic  
**c** Biotic **d** Biotic **e** Biotic  
**f** Abiotic **g** Abiotic  
**h** Abiotic **i** Biotic **j** Abiotic  
**k** Abiotic **l** Abiotic  
**m** Abiotic **n** Abiotic
- 2** Streamlined, muscles, fins, tail
- 3** Four legs, slim, muscular
- 4** Penguin, seal, turtle
- 5** To mate, rear young, escape from predators
- 6** Supported by water pressure
- 7** Community
- 8** **a** True **b** True
- 9** **a** To do with the land **b** To do with water
- 10** Layer of fat/blubber
- 11** Body heat is conducted away by water and temperature will fall
- 12** Birds: feathers, lay eggs, Mammals: skin/fur, live young, mammary glands
- 13** Chlorophyll
- 14** Yes, animals e.g. fish can live there
- 15** To see
- 16** Intensity of light decreases with depth. Hence ability to photosynthesise is lessened
- 17** Aerobic
- 18** CO<sub>2</sub>
- 19** C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>
- 20** Carbon dioxide plus water (and light and chlorophyll) produce oxygen and glucose
- 21** The temperature of water doesn't go up and down as much as it does on land

## 19 – Ecosystems

- 1** Producer
- 2** Secondary consumer
- 3** **a** Green plants **b** Bird of prey
- 4** No, too complex
- 5** The arrow means 'is eaten by' or 'energy is transferred to'
- 6** Three
- 7** Groups of individuals belonging to the same species

- 8** Sum of the different populations living in a given area
- 9** No. Digestive process is never 100% efficient
- 10** Step or strata in a food chain
- 11** Organism living off another creature without conferring any benefit, usually harming the host
- 12** Material wouldn't be recycled, less for other organisms to use
- 13** No, most important as nothing else could exist without the producers
- 14** **a** Producer **b** Primary consumer **c** Secondary consumer **d** Tertiary consumer
- 15** Each organism uses energy in e.g. moving, and other life activities
- 16** Teacher to check. Ratio should be 530 : 23 : 7 : 1

## 20 – The local environment

- 1** Temperature, wind
- 2** **a** To do with water **b** To do with the land **c** To do with trees
- 3** **a** Teacher to check **b** Teacher to check **c** Teacher to check
- 4** Teacher to check, but most are fairly close. Don't specifically include other living things
- 5** Seaweed
- 6** Ones that themselves are capable of breeding and producing offspring
- 7** Population: numbers of same species in an area  
Community: numbers of all organisms
- 8** **a** Where they are found  
**b** Variety
- 9** Amount of water, temperature, light, organisms present
- 10** Teacher to check. Include ways in which they interact with the abiotic as well
- 11** Teacher to check, e.g. some are predators, others not, etc.
- 12** Teacher to check
- 13** Teacher to check, e.g. certain desert plants won't survive the cold further south, etc.
- 14** As previous, e.g. roses grow well in hot areas but less well in tropical since they get black spot, etc.
- 15** Teacher to check

## 21 – Exobiology

- 1** Astrobiologists are concerned with life in the universe whereas astrophysicists are interested in the universe's components, living or otherwise



# Selected answers (cont'd)



- 2 It's all we've got to measure life by
- 3 Teacher to check, e.g. life evolved here and this can help us to understand how it might evolve elsewhere. Disadvantage is that life here evolved under conditions that may not apply elsewhere.
- 4 Light availability, temperature, etc.
- 5 Oil, pearls, wood, chalk, limestone, cotton, etc.
- 6 Like many investigations, a range of subjects is studied
- 7 If you're looking at a bacterium you'd need to focus on biology but the rock it may be found on may need a combination of geology and astro if it's from space. Some investigations will be more heavily oriented one way but still need the other.
- 8 Teacher to check for logical argument, e.g. some people argue that no new laws of science have been found, no new substances etc that didn't already exist in other fields, etc. It's often confused with sci fi's search for aliens, etc.
- 9 Teacher to check, e.g. vulcanism though as yet unproven, microbial
- 10 Water is required by cells, a good medium to grow things in
- 11 Stars that will act as suns. They must be large enough to burn long enough to give long-term, reliable heat
- 12 It will vary considerably, some places might have similar to that on Earth
- 13 Evolution of life in our universe. Or: Evolution of life in our universe or how planets and stars came about
- 14a Fossil evidence, oldest remains, etc.
- b Genetics, used to relate species, show where they branch off to form new ones
- c Gravity, force, mass, etc. How it affects life
- d How elements react to form compounds, how compounds react, etc.
- e Use knowledge of objects in the universe and how they resemble/differ from Earth. How one might affect the other
- 15 Teacher to check

## 22 – Ants

- 1 Genus, Species
- 2a Group of similar organisms that breed to produce fertile offspring
- b Only half the 'usual' number of chromosomes
- c Usual number of chromosomes, i.e. pairs
- d Unable to produce offspring through breeding
- e Feeds on dead animals and refuse
- f Eats other organisms (above producer level)
- g Special group/function
- 3 Teacher to check
- 4a 1.25 cm sec<sup>-1</sup> (approx.)
- b 1.4 cm sec<sup>-1</sup> (approx.)
- 5 Their ability to detect rain coming, nightfall, unforeseen presence of dead, dying or dropped food source other insects might detect, etc.
- 6 Teacher to check, e.g. the average temperature at which ants are most/least active
- 7 So that different species don't follow trail
- 8 Pheromone
- 9 Polymorphic, more than one shape/appearance, e.g. queen ant and worker ant look different

## 23 – Bacteria

- 1 Teacher to check
- 2 a 170 (approx.) b 340 (approx.) c 1500 (approx.) d 5000 (approx.) e 7500 (approx.)
- 3 No, will eventually run out of e.g. food/water
- 4 a Only one cell b More than one c Put into a group d Can only be seen using a microscope e Splitting into two f Without sex
- 5 Cause disease
- 6 Making antibiotics, preserving food
- 7 True

## 24 – Mouse plagues

- 1 Teacher to check
- 2 Mice can breed very rapidly. Good season, lots of food, followed by an event such as drought or disease

## 25 – Growth in offspring

- 1 Teacher to check
- 2 The bigger the size, the longer the gestation period
- 3 Body
- 4 Brain
- 5 Around 4 years
- 6 No, must eventually reach a maximum size
- 7 a 15 kg b 27 kg c 52 kg
- 8 Different diet, illness, not follow 'norm'

## 26 – How tall will I be?

- 1 162 cm
- 2 114 cm
- 3 185 cm
- 4 149 cm
- 5 176 cm
- 6 131 cm
- 7 Teacher to check
- 8 The greater the height of the father, the greater the height of the son
- 9 Height of mother, environmental factors
- 10 More sons/fathers, different age groups
- 11 Similar. Generally, taller mothers would have taller daughters. Their height would be, on average, less than the male equivalent. Males tend to be taller than females of the same age.
- 12 Improved diet, exercise and other lifestyle factors

## 27 – Antibiotics and disease

- 1 Between 1935 and 1945, numbers dropped
- 2 Between 1955 and 1965, numbers dropped dramatically
- 3 Bacteria – virus not affected by antibiotics
- 4 Teacher to check
- 5 Number of deaths increasing, antibiotics may be losing their effectiveness
- 6 Low grade fever, sore throat
- 7 35% of people who catch the disease die of it
- 8 87, 88
- 9 Infectious, catching
- 10 e.g. penicillin, streptomycin
- 11 Studies how often and where a disease occurs

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